DEPARTMENT OF ELECTRICAL AND INFORMATION ENGINEERING

FACULTY OF ENGINEERING
UNIVERSITY OF RUHUNA

INDUSTRIAL TRAINING REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE DEGREE OF THE BACHELOR OF THE SCIENCE OF ENGINEERING

ON

07TH NOVEMBER 2025

SAMSON INTERNATIONAL PLC
BOGAHAGODA, GALLE.

(04/08/2025 To 26/10/2025)

SEWINDA L.L.D. (EG/2021/4807)

PREFACE

After having a successful training at Samson International PLC, this report has been prepared to provide a fair understanding of the industrial training that I have undergone for 12 weeks, from 04th August 2025 to 26th October 2025.

During this second industrial training, I have learned to serve a vital bridge between theoretical knowledge and practical application, offering valuable exposure to real-world industrial environments. It provides opportunities to apply university-acquired knowledge and gain meaningful industry experience

This report consists of technical, management and social skills, knowledge, and experiences gained during the internship period. There are mainly six chapters. The first chapter provides the background of the organization, hierarchy, and related aspects of the company. The second chapter contains the technical working experiences obtained during the training period. Chapter three consists with the management experiences gained throughout the training period. Chapter four consists of the practice of Professional Standards and Engineering Ethics. Chapter five reflects on training experience giving considerations to environment and Sustainability. Finally, chapter six concludes with technical and management goals in an overall summary.

Sewinda L.L.D.

EG/2021/4807

Department of Electrical and Information Engineering,

University Ruhuna

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Department of Electrical and Information Engineering, University of Ruhuna, and National Apprentice and Industrial Training Authority (NAITA) for making all the necessary arrangements for making this training program successful and giving me this invaluable opportunity to work with Samson International PLC with the help of the guidance of the Coordinator of Engineering Education Center (EEC)

I would like to offer my deepest gratitude to Mr. Namal Nishantha (Assistant General manager), Mr. Nilantha J. Gamage (Manager - Engineering), Mr. Asanka Edirisinghe (Executive Production), who were my training supervisors for guiding me throughout the training period.

Finally, I should appreciate the support given by all the other staff members including Managers, Machine Operators, Supervisors and the workers to complete my three-month internship training program successfully.

Sewinda L.L.D.

EG/2021/4807

Department of Electrical and Information Engineering,

University Ruhuna

TABLE OF CONTENT

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

1. Introduction to the Training Establishment

1.1 Company Overview

1.1.1 Global history of Samson Group

Samson International PLC is a leading polymer-based compounding company in Sri Lanka, with ISO 9001:2008 accreditation. They are part of the DSI Samson Group of Companies, which comprises a diverse range of businesses including footwear, tires, food-grade rubber goods, trading, and information technology.

Figure 1.1: Mr. Samson Rajapaksa [1]

DSI was established in Galle in May 1962 - Samson Rajapaksa’s 50th birthday. Since the 1980s the expansion has been rapid. Showrooms have been set up in key towns. Factories have been modernized with state-of-the- art facilities. Local and international awards have been won. The product range expanded. DSI has not looked back and today there are 29 companies in the Group. Samson Rajapaksa’ s sons give the lead in managing the entire operation.

The second generation has strengthened the company making DSI a household name in Sri Lanka and capturing market leadership in the footwear industry. The third generation is gradually moving in to take the business to further heights. They have been able to convert the single business entity in to a conglomerate in keeping with the founder chairman's corporate philosophy. The DSI Samson Group, with over 60 years of experience is evolving itself to offer a vast range of products and services to the market.

1.1.2 The Samson Group in Sri Lanka [4]

- D. Samson Industries Ltd.
- D. Samson & Sons Ltd.
- Samson (exporters) Ltd.
- Samson Trading Company Ltd.
- Samson Rubber Industries (Pvt) Ltd.
- Samson Manufactures Ltd.
- Samson Rubber Products (Pvt) Ltd.
- Samson International Ltd.
- Samson Engineers Ltd.
- Samson Reclaim Rubbers Ltd.
- Samson Apparel Makers (Pvt) Ltd.
- Samson Compounds (Pvt) Ltd.
- Samson Sports wears (Pvt) Ltd.
- Samson Brush Manufactures (Pvt) Ltd.
- Vechenson (Pvt) Ltd.
- D. D. P. Packaging (Pvt) Ltd.
- Kelani Valley Canaries Ltd.
- D. S. R. Exporters (Pvt) Ltd.

1.1.3 History of Samson International PLC

Samson International PLC (SIL) is a well-recognized Sri Lankan company that has been manufacturing and exporting a wide range of rubber and PVC products since 1988. Listed on the Colombo Stock Exchange since 1992, SIL plays a key role in the country's industrial sector. As part of the DSI Samson Group, a respected business group with roots going back to 1962, SIL benefits from decades of experience, a strong operational structure, and access to valuable industry knowledge.

Quality has always been at the heart of SIL's operations. The company holds ISO 9001 certification, reflecting its commitment to maintaining international standards in both manufacturing and management. Its product range is broad and practical from household essentials like jar sealing rings and hot water bottles, to various types of rubber mats for home, industrial, and specialized uses. SIL also produces extruded rubber items like garden hoses, beadings, and automotive accessories such as mud flaps. They even manufacture custom rubber components designed to meet specific client needs.

With exports reaching over 30 countries, including markets in Europe, Asia, the Middle East, Africa, the Americas, Canada, Australia, and New Zealand, Samson International has built a strong global presence. This international reach reflects the company's ability to meet diverse market demands while consistently delivering high-quality products that meet global standards.

From its humble beginning in 1988 until becoming a front liner in the rubber industry, Samson International PLC has written a tale that is entrenched with excellence during its quest to be positioned as a leader in rubber manufacturing in Sri Lanka.

- 1988 - Incorporation
- 1992 - listed in Colombo stock Exchange
- 1994 - Became the first rubber manufacturing company in Sri Lanka to obtain ISO 9001 certification
- 1995 - Started manufacturing rubber ho5 water bottles with British standard and TUV certification
- 2007 - invested in the first microwave continues vulcanizing machine in Sri Lanka
- 2009 - Acquired all assets of Aksel (Pvt) Ltd in Kalutara
- 2011 - Recognized as the best value-added rubber product exporter in Sri Lanka for the first time at National Export Awards
- 2013 - celebrated its silver jubilee
- 2014 - Acquired all assets of Okta PVC Lanka (Pvt) Ltd which manufacture PCV products.
- 2015 - Started manufacturing products with FSC certification
- 2016 - Received Business Social Compliance Initiative (BSCI) certification
- 2017 - Received ISO 14001 certification for Environmental Management Systems. Received ISO 50001 certification for Energy Management Systems
- 2020 - Recognized as the best value-added rubber product exporter in Sri Lanka for the 6th consecutive year at the NCE export awards. [4]

1.1.4 Compliances of Samson International PLC.

Table 0.1: Compliances of Samson International PLC [4]

Compliances	Logo of the Compliances
REACH (Registration, Evaluation, Authorization and Restriction of Chemicals)	
FSC Certification	
ISO 14001: 2015 Environmental System Certification	
Energy Management System Certification ISO: 50001	
ISO 9001: 2015 Quality system Management	
BSCI (Business Social Compliance Initiative)	

1.1.5 Company Profile

- Company Logo:

Figure 1.2: Company Logo [3]

- Company Name: Samson International PLC
- Company Short name: SIL
- Address: Samson International PLC, Akuressa Road, Bogahagoda, Galle.
- Website: https://www.samsonint.com

1.1.6 Samson International Global Reach

No. 1 rubber product manufacturer in Sri Lanka. We are proud to be the reliable supplier of rubber products for leading distributors and manufactures in over 30 different countries around the world. At SIL, our approach to serving our clients is driven by the determination to work together across boundaries to leverage our collective expertise and help our business and our clients succeed.

Asia

- Bangladesh
- China
- India
- Japan
- Maldives
- Pakistan
- Saudi Arabia
- South Korea
- Turkey

Europe

- Germany
- Lithuania
- Netherlands
- Norway
- Poland
- Romania
- Sweden
- Switzerland
- United Kingdom

1.1.7 Products of Samson International

1. Pharmaceutical Industry
    - Hot water Bottles
    - Menstrual Cups

2. Food packaging Industry
    - Jar Sealing Rings
    - Bottle Caps

3. Bath ware Industry
    - Bathmats
    - Soap holders

4. Automotive Industry
    - Three-wheeler mats
    - Scooter mats
    - Mud flaps
    - Car carpets

5. Matting and Flooring Industry
    - Floor mats
    - Rubber Floor tiles
    - Floor carpets

6 Foot ware Industry

- Shoe soles
- V strap for waves slippers
- Rubber slippers

7 Construction Industry

- Samson PVC pipes
- Wire casings
- Rubber beadings
- Road bumpers

8 Irrigation and Agricultural Industry
- Irrigation Pipes
- Flower Pots
- Seed trays

9. Other Industries
    - Rubber balls
    - Tennis balls
    - Erasers

Figure 1.3: Pharmaceutical Products of Samson International [4]

Figure 1.4: Food and Packaging Products of Samson International [4]

Figure 1.5:Automotive Products of Samson International [4]

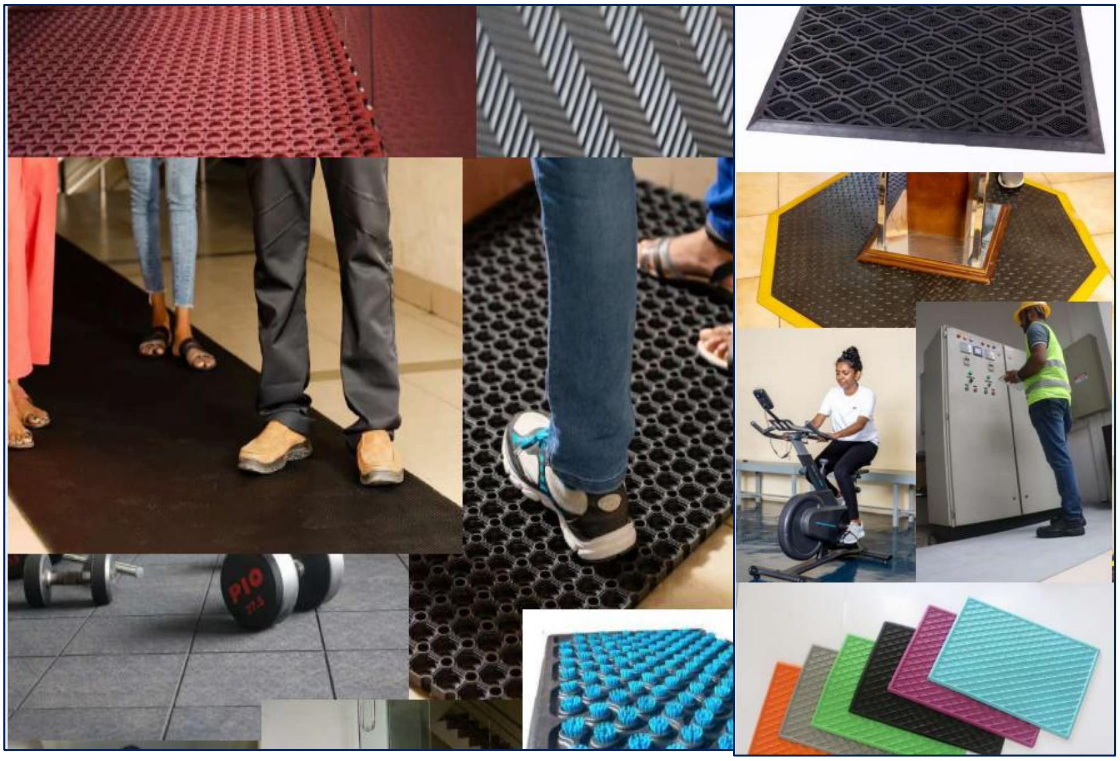


Figure 1.6:Mats and Flooring Products of Samson [4]

Figure 1.7:Foot ware Products of Samson International [4]

Figure 1.8:Construction Industry Products of Samson International [4]

Figure 1.9:Other Products of Samson International [4]

1.2 Vision and Mission

1.2.1 Vision

To be the leading technology company in the region for polymer-based compounds and products.

1.2.2 Mission

To become a multi-skilled manufacturer by providing maximum value to our business partners by utilizing modern technology to the fullest extent and offering diverse polymer-based products to the global market.

1.2.3 Values

- Contributing to the socio-economic development of the country.
- Respect for our national and religious heritage.
- Conservation and protection of the environment.
- Development of disciplined human resources based on family concept.

1.3 Organization Charts

D. Samson Industry is a family-owned business in Sri Lanka. so, when we consider its organization chart, its top-level position is maintained by family owners. "The Chairman, Group Manager, and Plant Manager" are the three top positions.

Figure 1.10: Organizational chart of samson group

Samson International PLC is governed by a Board of Directors comprising seven members. Board is chiefly responsible for monitoring managerial performance while guiding corporate strategy and achieving an adequate return for shareholders. The main task of the Board of Samson International PLC is to set the company's direction and strategy. They concentrate on directing the Company and not on managing it. They direct to guide managers the way to be ahead and lead. The Board balance the review of past performance and forward-looking discussion. The Board also limits the time spent on routine & administrative matters. They are expected to prevent any conflict of interest and balance competing demands of the company. Managers manage the company. Managing is about dealing, hands-on with the design, implementation and maintenance of effective operating models and prudent control systems.

The functions of the Board of the Samson International Plc include the following.

- Obligation - Ensure the company’s well-being
- Responsibility - Safeguard stakeholder interest
- Entrepreneurship - Direct the company’s affairs
- Trustee - Satisfy shareholders’ interest
- Effective control - Guide and control executive management
- Compliance - Ensure that legal, ethical, and moral obligations are met

Our directors are aware that they have a duty of care as well as fiduciary duties. They are required to exercise this duty of degree of care, diligence and skill reasonably and do not undertake their legitimate duties to perform negligently. In the case of fiduciary duties, directors are in position of trust in relation to the Company and act in faithful, trustworthy manner towards or the company’s behalf.

This is the Director board of samson International PLC.

- Dr. D. Kulatunga Rajapaksa - Chairman
- Mr. Tissa K. Bandaranayake - Independent, Non–Executive Director
- Mr. D. G. Priyantha S. Abeygunawardana - Director / General Manager
- Mr.D. Dilshan A. Rajapaksa - Managing Director
- Mr. D. Chandula J. Rajapaksa - Non–Executive Director
- Ms. Indira Malwatte - Independent, non–Executive Director
- Mr. D. Nishan S. Rajapaksa - Non–Executive Director

From this Director board 3 members are Executive Directors.

- Dr. D. Kulatunga Rajapaksa - Chairman
- Mr. D. G. Priyantha S. Abeygunawardana - Director / General Manager
- Mr.D. Dilshan A. Rajapaksa - Managing Director

Executive management takes to the highest level of leadership within an organization. Our executive management team is responsible for setting strategic direction, making key

decisions, and overseeing the overall operations and performance of the organization. They work closely with the board of directors to develop and implement strategies, allocate resources, manage risks, and drive the organization toward its goals and objectives. Additionally, executive management often plays a critical role in representing the organization to external stakeholders, including investors, customers, partners, regulators, and the public.

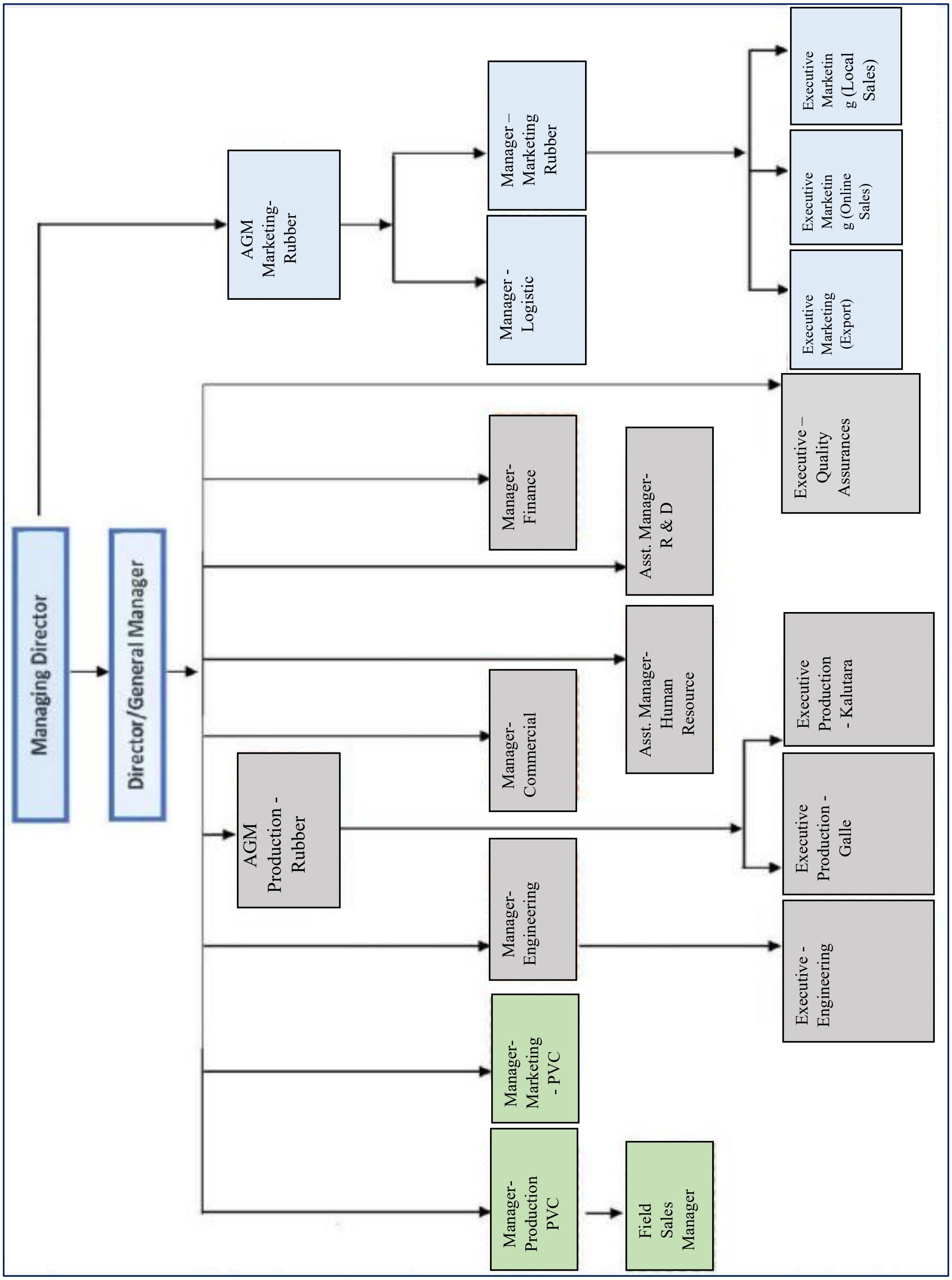


Figure 1.11: Organization chart of samson International PLC [4]

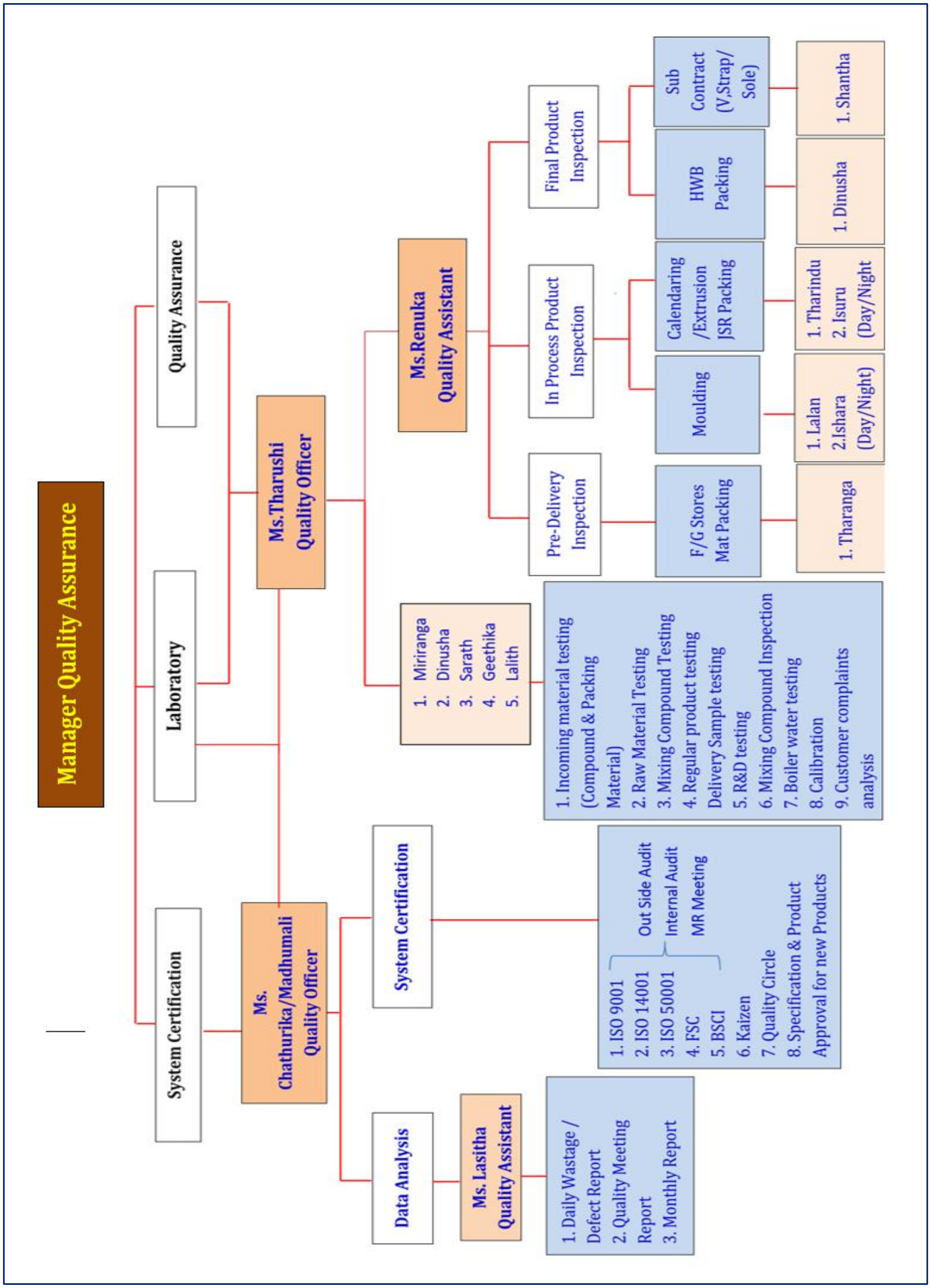


Figure 1.12: Organization chart of Quality Assurance Department SIL [4]

CHAPTER 2

2. Training Experiences - Technical

This chapter consists with the technical experience and the practical knowledge that was obtained from the 12 weeks of training at Samson International PLC. I was assigned to the Moulding Section under the supervision of Mr. Asanka Edrisinghe. In there, Mr. Asanka wanted to have a real-time monitoring system for several machines to improve labor efficiency and production count as well. My main project was to give a full IoT based solution model for this. In the first two to three weeks, Mr. Nama Nishantha (AGM) and the other supervisors visited all sections to find optimal solutions to minimize manpower and improve efficiency. So, in that period, I was assigned to improve the JSR Packing. (Counting the various types of food-grade jar sealing rings). The other two interns(mechanical) and I designed a machine learning model for counting the rings. That project was a joint project and got some exposure to machine learning as well.

2.1 Introduction to the IoT based machine Monitoring and Recording Project

2.1.1 Project Overview

At the time, the moulding section was in full operation, primarily focused on the production of rubber bathmats, and various types of carpets. While the machinery was robust, the production monitoring and output counting processes were largely manual. Foremen and operators relied on physical logs and manual tallies to record hourly production. Some older machines, lacked any digital interface for real-time tracking. My assignment began exactly when the management want to transition from "Wait & Fix" manual reporting to a digital, real-time "Predict & Prevent" monitoring framework. Here I was able to implement the solutions for the following machines of the moulding section,

- E – 69: Road bumpers/ Flower pots
- E – 125: Bathmats
- E – 94: Mud carpets/ Floor carpets

Solution: Development of a scalable IoT system using ESP32 microcontrollers and the ThingsBoard platform to monitor machine cycles, visualize hourly counts, and generate production reports.

2.1.2 Project Scope

1. **Pilot Implementation:** Developing a prototype for the E-25 Press machine (for testing the full solution with the minimum dashboard end options)
2. **Scaling & Retrofitting:** Expanding the system to older machines (E-69) and new machines (E-103, E-94).
3. **Hardware Design:** Designing PCBs and industrial enclosures for harsh factory environments.
4. **Dashboard Development:** Creating dashboards and assign them for supervisors.

2.1.3 System Architecture and Technologies

System Architecture and Component Selection

To achieve a scalable and cost-effective solution, designed an IoT architecture based on the ESP32 microcontroller and the ThingsBoard - demo platform which was free to implement the projects with some limitations. The selection of components was driven by specific technical requirements and environmental constraints.

<u>Microcontroller: ESP32-WROOM-32</u>

Selected the ESP32 Development Board as the core controller for the following reasons:

- Integrated Wi-Fi & Bluetooth: Unlike the Arduino Uno, the ESP32 has a built-in Wi-Fi stack, eliminating the need for external modules like the ESP8266, which reduces circuit complexity and failure points.
- Inbuilt Noise Handling Mechanisms: Helped to set the input pull-up This was crucial for preventing missed counts during network reconnection/noise attempts.
- 3.3V Logic with 5V Tolerance: While primarily a 3.3V device, it interfaces well with industrial sensors when using level shifters.
- Cost-Efficiency: At approximately LKR 1350-1650, it fit within the low-cost budget requirement for scaling to multiple machines.

Signal Acquisition Sensors

The choice of sensor depended entirely on the generation of the machine (Old vs. Modern).

<u>A: For Old Machines (E-69, E-25): Mechanical Relays</u>

- Context: Older machines like the E-69 did not have digital outputs. However, they utilized a high-voltage (230V AC) timer to control the "Curing Time" of the rubber product.
- Component: 230V AC to 5V DC Relay Module.
- Reasoning: I needed a way to electrically isolate the machine's high-voltage logic from my low-voltage microcontroller. By tapping into the timer's buzzer output, the relay would trigger only when a cycle was successfully completed, providing an input signal.

<u>B: For New Machines (E-103): Inductive Proximity Sensors (NPN)</u>

- Context: Non-Intrusive Integration for Modern PLC Machines **(E-103 & E-94)** The E-103 bathmat press and the E-94 press are modern machines controlled by Programmable Logic Controllers (PLCs). As these machines were still within their manufacturer's warranty period, I did not have permission to interface directly with their internal PLC inputs or modify the existing wiring.

To address this constraint, I designed a standalone monitoring device that operates independently of the machine's internal control system. Instead of tapping into internal signals, I utilized Inductive Proximity Sensors. These sensors were mechanically mounted on the press frame to detect the mould movements externally. This approach ensured the monitoring system remained electrically isolated from the machine, preserving the warranty while strictly fulfilling the requirement for a separate, non-intrusive counting mechanism.

Figure 2.1: Proximity Sensor's Placement to Detect the Moving Metal

- Component: Inductive Proximity Sensor SN04-N NPN (ON).
- Reasoning:
    - *Non-Contact Detection:* These sensors detect the metal surface of the mould as it closes, without physical wear and tear.

- *NPN Configuration:* I chose NPN (Normally Open) because it acts as a "Low-Side Switch." When the mould closes, the sensor connects the signal pin to Ground. This is safer for the ESP32 as we can use the internal pull-up resistors or a simple voltage divider, reducing the risk of accidental high-voltage injection compared to PNP sensors.

Voltage Regulation & Protection

- **Logic Level Shifter (4-Channel):** Industrial sensors and relays often operate at 5V. The ESP32 GPIOs are 3.3V. I used a bi-directional logic level shifter to safely step down the 5V signals to 3.3V, protecting the MCU inputs.
- **Industrial Enclosure (IP Box):** A $1000 \times 700 \times 600$ mm (scaled to device size) IP-rated enclosure was selected to house the components. This was essential to protect the electronics from conductive carbon dust and oil mist present in the moulding section.

Software Stack

- Firmware Development: VS Code with PlatformIO was used for programming the ESP32 in C++.
- Communication Protocol: MQTT (Message Queuing Telemetry Transport) was chosen for its lightweight nature, perfect for communication in environments with potential network instability.
- IoT Platform: ThingsBoard (Community Edition).
    - Used for data visualization (Widgets, Gauges).
    - Device management and authentication via Access Tokens.
    - Day to day system updating.
    - Rule Chains for server-side processing of "Day Counts" and "Night Counts".

2.1.4 Circuit Design for three machines.

In here, I distributed mainly into two phases for each machine since there were three circuits for each machine based on their unique requirements.

Phase 01: For E – 69 machine

The E-69 is an old machine, and its architecture required a specialized approach to integrate it with the mechanical relay system. The development of the circuit for this machine was based on the relay triggering after the cycle time (curing and the settling time of the compound),

Figure 2.2: E - 69 Press Machine

Design Challenge: Interfacing high-voltage AC signals with a 3.3V microcontroller. Solution: I used a 230V AC Relay since it can handle the dc switching as well by giving 5V input to trigger the relay.

1. Input: The machine's timer signal (230V) triggers the relay coil.
2. Output: The relay switches a 5V DC signal.

Level Shifting: The 5V signal is passed through a Logic Level Shifter to step it down to 3.3V before entering the ESP32 GPIO pin (GPIO 4)

Easy EDA Circuit Design

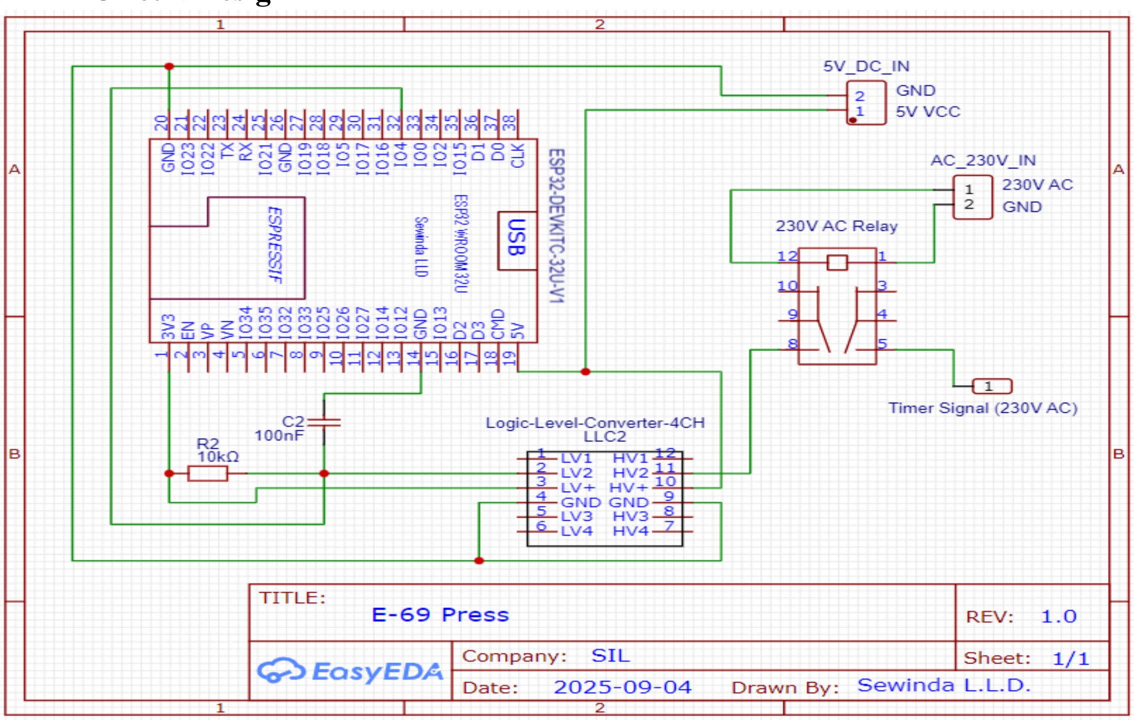


Figure 2.3: Easy EDA Circuit Design for E - 69 Press Machine

The initial circuit design proved susceptible to electromagnetic interference and mechanical noise, which led to signal instability. To ensure accurate data collection, a two-tiered solution involving both physical signal filtering and software optimization was implemented.

1. **Physical Filter Implementation (Hardware)**: To mitigate noise at the source, a low-pass RC filter was physically installed on the signal line. This arrangement utilizes a $10k\Omega$ resistor and a $100\,nF$ capacitor to smooth the low-voltage signal. This hardware modification effectively filters out high-frequency noise spikes, preventing erroneous counting and ensuring a clean signal reaches the microcontroller.

2. Firmware Optimization (Technical Fix): Complementing the hardware filter, the firmware on the ESP32 was updated to enhance signal reliability:

- Internal Pull-up: The code was modified to enable the ESP32's internal pull-up resistors, ensuring the input pins remain in a known state (HIGH) when no signal is present.
- Software Debounce: A specific "debounce" delay algorithm was introduced to ignore transient signal fluctuations.
- Logic Restructuring: The timer connectivity logic was rearranged to strictly distinguish valid timer signals from spurious triggers caused by the machine's mechanical vibrations.

Example:

```
// Initialize button pin
  pinMode(BUTTON_PIN, INPUT_PULLUP);
  Serial.println("Button initialized on pin " + String(BUTTON_PIN));
-unsigned long last_button_time = 0; // Track last valid button press time (for x-second prevention)
unsigned long button_debounce = 350; // 200ms debounce for industrial button
```

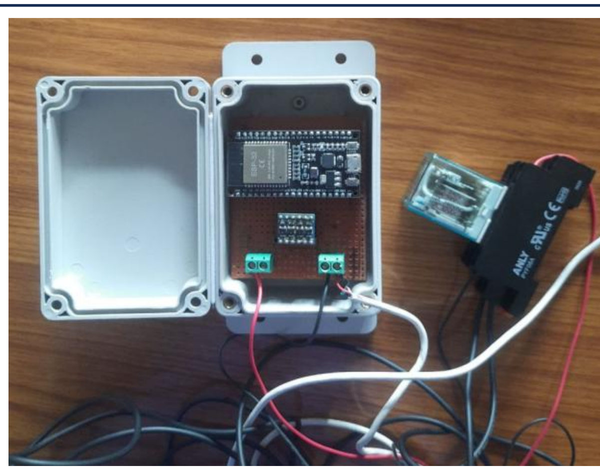


Figure 2.4: E - 69 Dot Board Circuit Before Fixing to the Machine

Phase 02: For E – 103 machine and E -94 machine

Figure 2.5: E - 103 Press Machine

The remaining two units are modern PLC-based automated machines. Unlike the legacy E-69 model, these machines did not feature an accessible internal timer circuit that could be tapped for data extraction. Therefore, an external sensing solution was required to monitor production cycles without interfering with the machine's internal logic.

Selected Solution: Inductive Proximity Sensors

To capture the machine cycle externally, I selected the SN04-N NPN (NO) inductive proximity sensor. This sensor was chosen for its reliability in industrial environments and its compatibility with the previously established noise cancellation and power regulation mechanisms.

Implementation Details:

- **Mounting Strategy:** The sensor was rigidly mounted on the machine frame, positioned specifically to detect the metal mould when it reaches the **bottom position** of its movement cycle. There was a valid time duration based on the user input to get the cycle count most accurately which will cover in the dashboard part. This physical position correlates with one completed unit of production.
- **Electrical Characteristics:** The sensor operates within a **6V–36V DC** range. This wide voltage tolerance made it easy to integrate into the existing power setup while maintaining sufficient sensing distance.

- **Signal Logic (Active Low):** Since the sensor is an **NPN** type, the signal line floats High when idle. When the metal mould is detected:
    1. The sensor connects the signal line to **Ground (Low)**.
    2. The ESP32 detects this transition.
    3. The firmware is configured to trigger an interrupt on the **FALLING edge** of the signal, ensuring precise cycle counting.

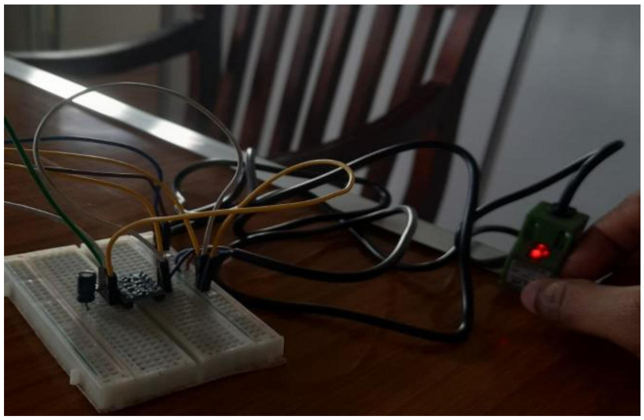


Figure 2.6: Breadboard testing the Circuit

Easy EDA Circuit Design

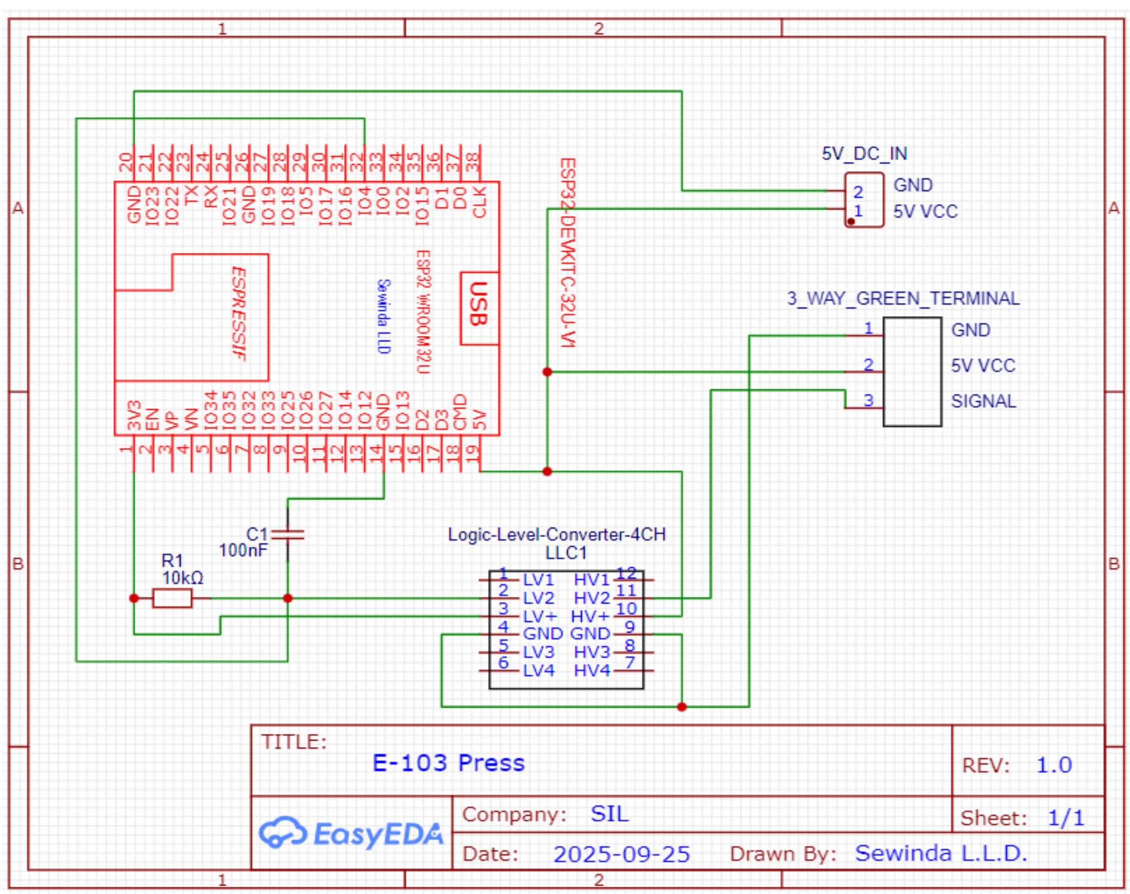


Figure 2.7: Easy EDA Circuit design for the E -103 Press Machine

*Note: The circuit design for the E-94 Press Machine mirrors the previous setups but incorporates an additional manual input button to handle specific production anomalies. Since this machine frequently switches its moulds, the occasional use of lower-cost rubber compounds for local markets can result in curing damage, requiring operators to re-cure the same product by adding some raw compound. Because this secondary process would typically trigger a false production count, the manual button was integrated to signal repair cycles; the system cross-references this input with the valid time durations set in the dashboard to strictly prevent the operator's mis-input signals. The valid time durations are configured on the dashboard for each specific mould and product type. This customization is necessary because curing times vary significantly between different orders, ensuring the validation logic matches the specific production requirements.

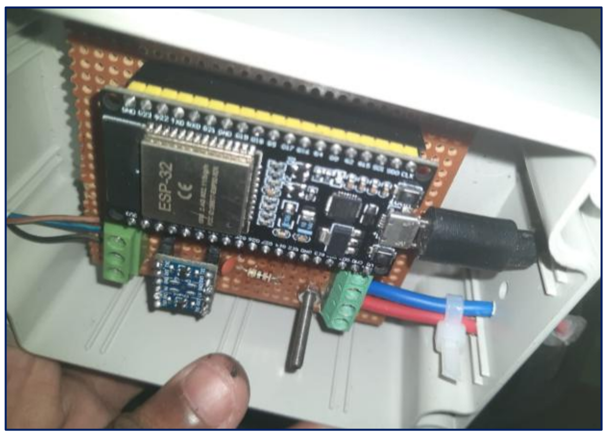


Figure 2.8: Implemented Circuit for the E - 103 Press Machine

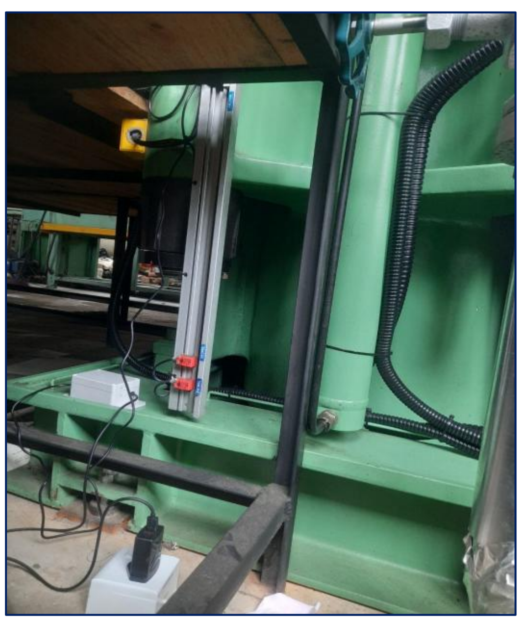


Figure 2.9:Setup and Testing of the Solution

Component List Purchased

1. ESP32 Microcontroller
2. 1000×700 mm Industrial IP box with hanging
3. Dot Board (Specially check)
4. Logic Level Shifter
5. SN04-N NPN NO Inductive Proximity Sensor
6. Relay Module + Base
7. Green Terminal Connectors
8. AC-DC Power Adapter 230V to 5V
9. USB Cable
10. DC Circuit Wires + Pin Headers (Female)

Figure 2.10: SN04-N NPN NO Inductive Proximity Sensor [5]

Figure 2.11: Industrial Relay Module + Base [6]

2.1.5 Dashboard Designing [2]

A core requirement of the project was to digitize a legacy industrial press machines (E-103, E-69, E-94) by implementing a real-time monitoring system. The core requirement was to track production counts, remotely update, manage shift-based data (Day/Night), and record "damage" or rejected units. The solution needed to be robust, allowing for wireless configuration and remote monitoring without physical intervention at the machine level. I utilized ThingsBoard Demo (Community Edition) for this purpose. Below is the technical flow of how the dashboard environment was constructed

Why Thingsboard:

After evaluating several IoT platforms (including Blynk, AWS IoT Core, and custom web servers), **ThingsBoard** was selected as the dashboarding solution for the following reasons:

- **Industrial Standard Widgets:** Unlike hobbyist platforms, ThingsBoard offers professional-grade widgets (Digital Gauges, Timeseries Charts, Control Knobs) suitable for a factory floor environment.
- **RPC (Remote Procedure Call) Capabilities:** The critical requirement to *send* data back to the machine (e.g., setting Op_time or Moulds count) is natively supported via Two-Way RPC.
- **Rule Chains:** Complex logic (such as alarm triggers or data transformation) can be handled server-side.
- **Data Retention:** Built-in SQL database handling for historical data analysis (Timeseries) allows for long-term production trending.
- **MQTT Support:** Native support for the lightweight MQTT protocol ensures low-latency communication with the ESP32 hardware.
- **Thingsboard Live:** This is a mobile app accessible to anyone with an account. If a user is assigned to a specific dashboard, they can view it from the mobile app as well.

Step 1: Device Provisioning

1. **Login:** Accessed the ThingsBoard instance via the web interface using the email.

2. **Create Device Profile:** “Moulding Section” As a this is separate profile any device in that particular section can recognize uniquely. If this scale, other devices (Machines) can add to this profile.
3. **Create Devices:** Navigated to the "Devices" section and added a new device and assign them to the “Moulding Section” profile. (Names of the devices: E_69, E_103, E_94…)
4. **Credentials:** Generated a unique Access Token for each device. This token was hardcoded into the ESP32 firmware *(#define TOKEN "YOUR_ACCESS_TOKEN")* to authenticate MQTT messages.

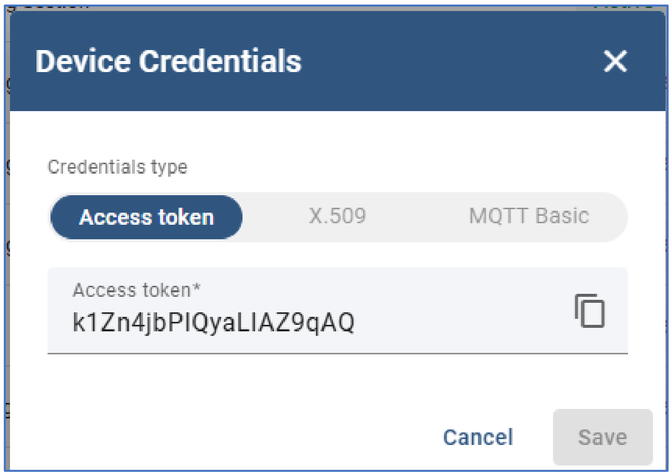


Figure 2.12: E -103 Device Access Token

5. **Creating Entity Alias:** Created an Entity Alias named as moulding_section. This was helpful if we need to extend the same machine functions for different machine. (Ex: E – 69 machine functions to the E -103 machine functionalities just adding only new device)

Step 2: Attributes Configuration& Telemetry

In the ThingsBoard platform, Attributes are key-value pairs used to store static or semi-static data about a device. Unlike telemetry (which is time-series data like total_count), attributes represent the current *state* or *configuration* of the machine.

For an example: E-69 Press, three specific scopes of attributes were utilized to manage the data flow between the edge device (esp32) and the cloud dashboard.

1. Server Attributes

Server attributes are strictly server-side variables. They are used by the ThingsBoard Rule Engine or the system administrator to manage the device's lifecycle. These values are never sent to the physical device; they exist only in the cloud to monitor connectivity and health.

List of Attributes (from System Logs):

- **Active**: Boolean flag indicating if the device is currently considered "active" by the rule chain.
- **InactivityAlarmTime:** Timestamp used to trigger an alarm if the machine stops communicating for too long.
- **LastActivityTime**: The exact Unix timestamp of the last message received from the device.
- **LastConnectTime**: Timestamp of the last successful MQTT connection.
- **LastDisconnectTime**: Timestamp of when the device last went offline.

These attributes are used to drive the "Online/Offline" status widget on the dashboard. The dashboard checks LastActivityTime to determine if the machine is live.

2. Shared Attributes

Shared attributes are dynamic configuration settings. They serve as the primary control mechanism for the dashboard operator to change machine settings remotely.

- Direction: Dashboard (Server) → Device (Client).
- Mechanism: When an operator changes these values on the dashboard, ThingsBoard sends an update notification to the ESP32. The ESP32 then updates its internal logic (e.g., changing the mould multiplier. Ex: moulds_Day).

  List of Attributes (Configuration Parameters):

- **moulds_Day**: (Integer) Defines the number of active moulds for the Day Shift (6 AM - 6 PM).

- **moulds_Night**: (Integer) Defines the number of active moulds for the Night Shift.
- **Op_time** (implied): The target operation cycle time (in seconds) used for validation.
- **day_assignee / night_assignee**: (String) The names of the supervisors assigned to the respective shifts.
- **target**: (Integer) The daily production goal.
- **day_correction / night_correction**: (Integer) Values used to manually adjust counts in case of errors.
- **Colour_code**: (String) Metadata for the specific product being manufactured (e.g., "white").
- **comments:** (String) Maintenance notes or operator logs entered from the dashboard.
- **day_correction / night_correction**: To update the count, if any maintenance or sample checking happens.

These are vital for the Remote Procedure Call (RPC) system. For example, if the mould count changes from 2 to 4, the operator updates moulds_Day in the Shared Attributes, and the ESP32 automatically recalculates production counts without requiring a firmware re-upload.

3. Client Attributes

Client attributes are read-only data points reported by the device to the server. They provide visibility into the ESP32's internal state, firmware status, and hardware health.

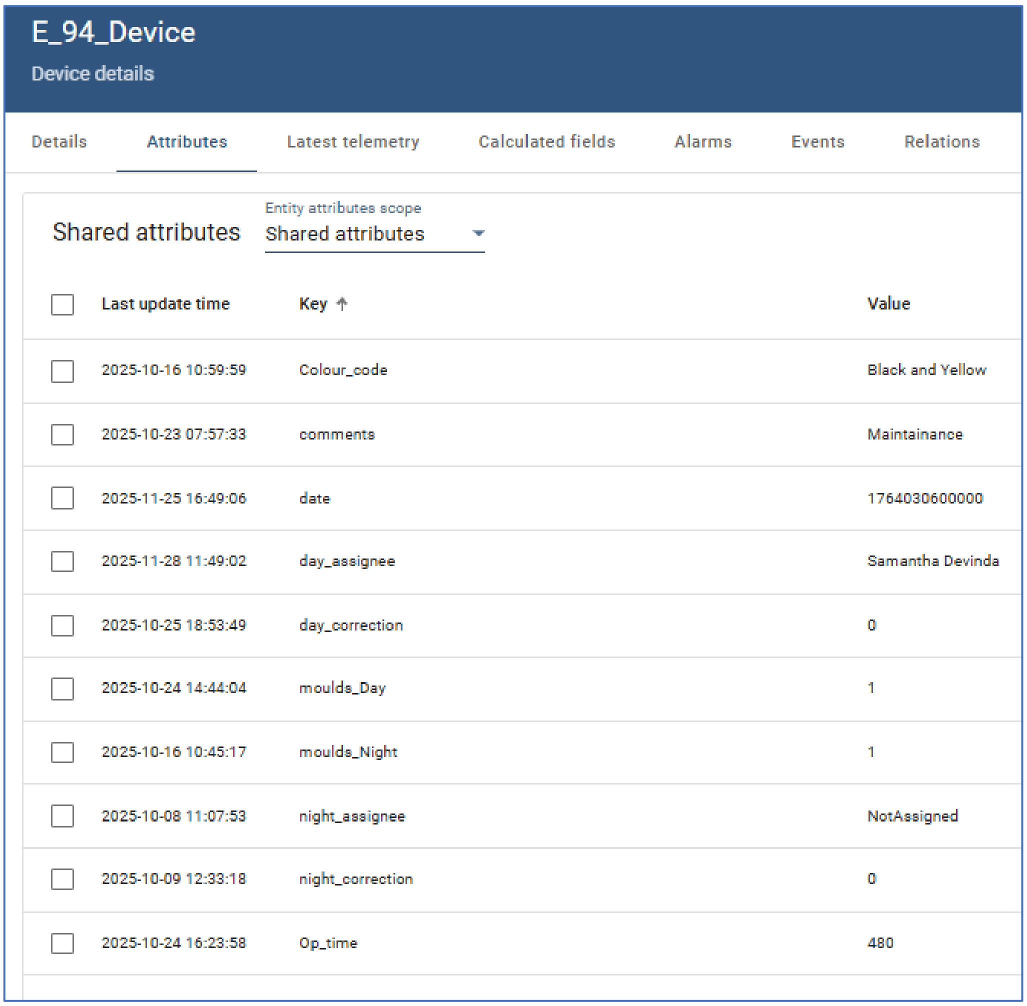
- Direction: Device (Client) →Dashboard (Server).
- Mechanism: The ESP32 publishes these values on startup or when a state change occurs.

List of Attributes (Device State):

- **boot_time**: Timestamp of the last ESP32 restart. Crucial for detecting power failures.
- **activity_state**: Current operational mode (e.g., "RUNNING").
- **connection_status**: Reports "connected" when the device successfully links to WiFi/MQTT.
- **connection_lost**: Boolean flag indicating if a connection drop occurred recently.

- **current_date**: The date currently stored in the ESP32's memory (used to verify NTP sync).

These attributes allow for remote debugging. by looking at boot_time and connection_status in the "Client Attributes" tab, designee can diagnose if a data gap was caused by a power outage or a network failure without visiting the factory floor.



Last update time	Key	Value
2025-10-16 10:59:59	Colour_code	Black and Yellow
2025-10-23 07:57:33	comments	Maintainance
2025-11-25 16:49:06	date	1764030600000
2025-11-28 11:49:02	day_assignee	Samantha Devinda
2025-10-25 18:53:49	day_correction	0
2025-10-24 14:44:04	moulds_Day	1
2025-10-16 10:45:17	moulds_Night	1
2025-10-08 11:07:53	night_assignee	NotAssigned
2025-10-09 12:33:18	night_correction	0
2025-10-24 16:23:58	Op_time	480

Figure 2.13: Some of Shared Attributes of E-103 Press machine

In this system, **Telemetry** refers to the dynamic, time-series data stream transmitted from the ESP32 to the ThingsBoard cloud to track real-time operational performance. Unlike attributes which represent static configuration settings, telemetry data is indexed by a timestamp upon receipt, allowing the platform to build a historical record of the machine's output. Key telemetry variables implemented in this project include day_count and night_count for monitoring shift-specific progress, total_count for cumulative lifetime tracking, and damage_dash for logging rejected units. This data feeds directly into the dashboard's digital gauges and timeseries charts, enabling management to not only view the immediate status of the press machines but also to analyze production trends and efficiency fluctuations over days, weeks, or months.

Step 3: Dashboard & Widget Creation

The dashboard design follows command and control method, divided into three areas,

1. Input and Control Widgets: Machine, operator identity and initial configurations for a particular day (Static/Semi-static).
2. Real Time Widgets: Real-time production counts with validations (Esp 32 telemetry).
3. Analylical widgets: Historical data visualization and logs maintain (Timeseries Data).

<u>Input & Control Widgets (RPC Driven)</u>

These widgets are critical for making the system dynamic. Instead of hard-coding values like "Mould Count" or "Target" in the firmware, these widgets allow the floor supervisor to configure the machine wirelessly.

Selected Widgets:

- **Moulds_Day / Moulds_Night:** Allows the operator to define how many cavities are active (1-4)
- **Maximum Operation Time (Op_time):** For a particular product what is the product cycle time.
- **Target:** Sets the daily goal (e.g., 225 units). This enables conditional formatting
- **Assignee Inputs (Day/Night):** Digital accountability. It logs who was responsible for the shift, replacing manual paper logbooks.
- **Comments, Day/Night correction:** To update if any miss count happens.

<u>Real-Time Telemetry Widgets</u>

These widgets are designed for instant updating monitoring. They update immediately to reflect the current count of the press machine.

Selected Widgets:

- **Total Count (Big Number):** The count is the most important metric. Choose a large, bold font widget so it is readable from a distance on the factory floor monitor.
- **Day/Night Count:** Shift-specific tracking. These reset automatically at 6:00 AM/PM, giving instant feedback on the current shift's performance.
- **Device Status (LED Indicator):** A simple Green/Red indicator linked to the active server attribute. It tells the operator immediately if the system has lost connectivity with the dashboard.
- **TimeStamp:** Displays the last valid sync time. This builds trust with the operator, confirming that the data displayed is live.

<u>Analytical Widgets (Timeseries & History)</u>

These widgets are for the manager/engineer. They visualize trends rather than just current values.

Selected Widgets:

- **Cumulative Hourly Count (Line Chart):** This sawtooth wave pattern clearly shows production rhythm. Steep slopes indicate fast production; flat lines indicate idle time (breakdowns or breaks). It provides a visual "heartbeat" of the machine.
- **History Table:** It logs every single telemetry update (Timestamp, Counts, Assignee). This is crucial for disputes (e.g., "Was the machine running at 11:30 PM?").
- **Corrections & Damage Tables:** Transparency. Since operators can manually edit counts (using the Correction widgets), these dedicated tables provide an audit trail of who changed a count when and what reason.
- **Damage or Sample Updating Table:** This indicates if any damage occurred or if sample checking took place. Sometimes, several samples are checked before the whole order proceeds.

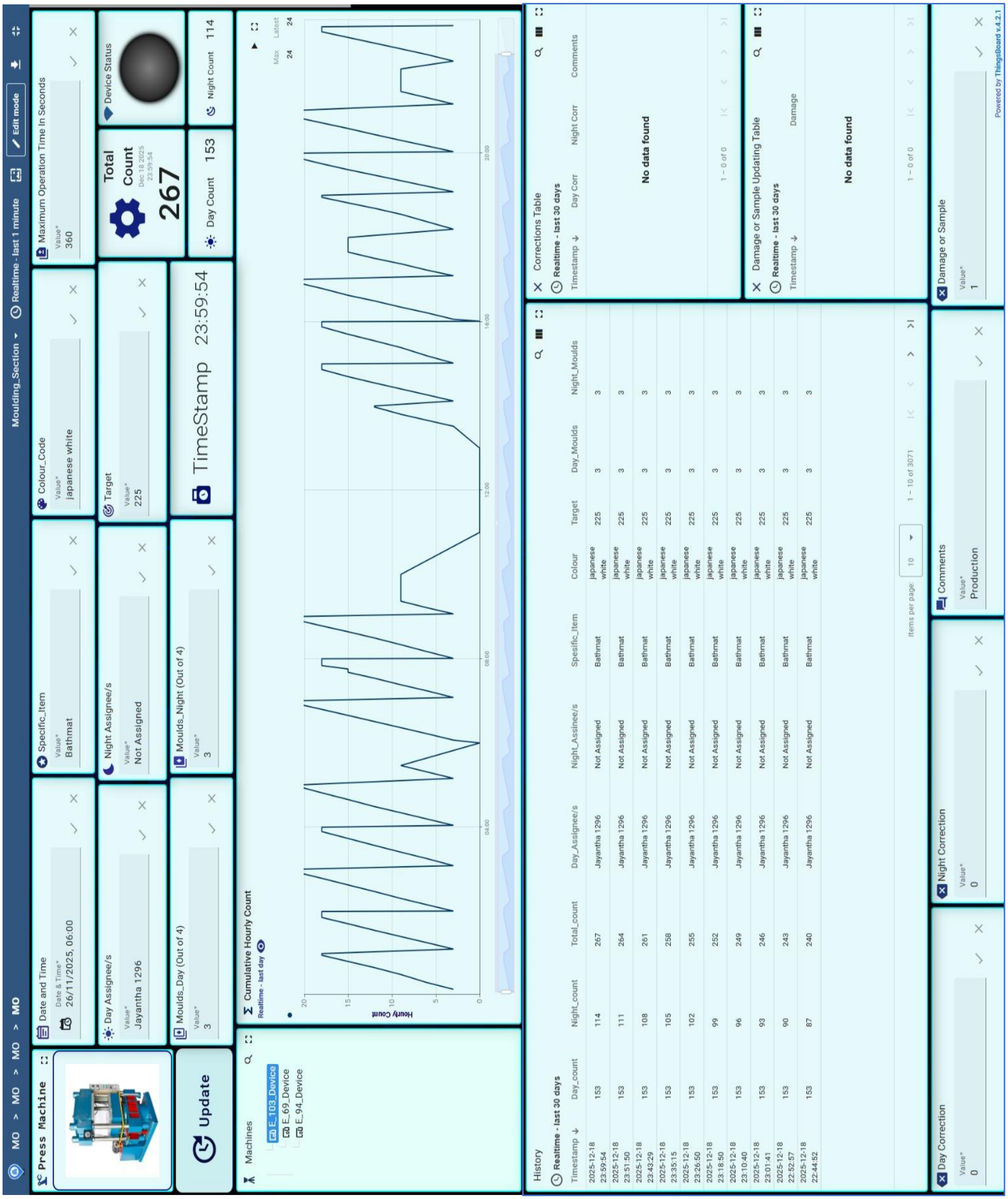


Figure 2.14: E-103 Press Machine Dashboard

Step 4: Rule Chain Handling

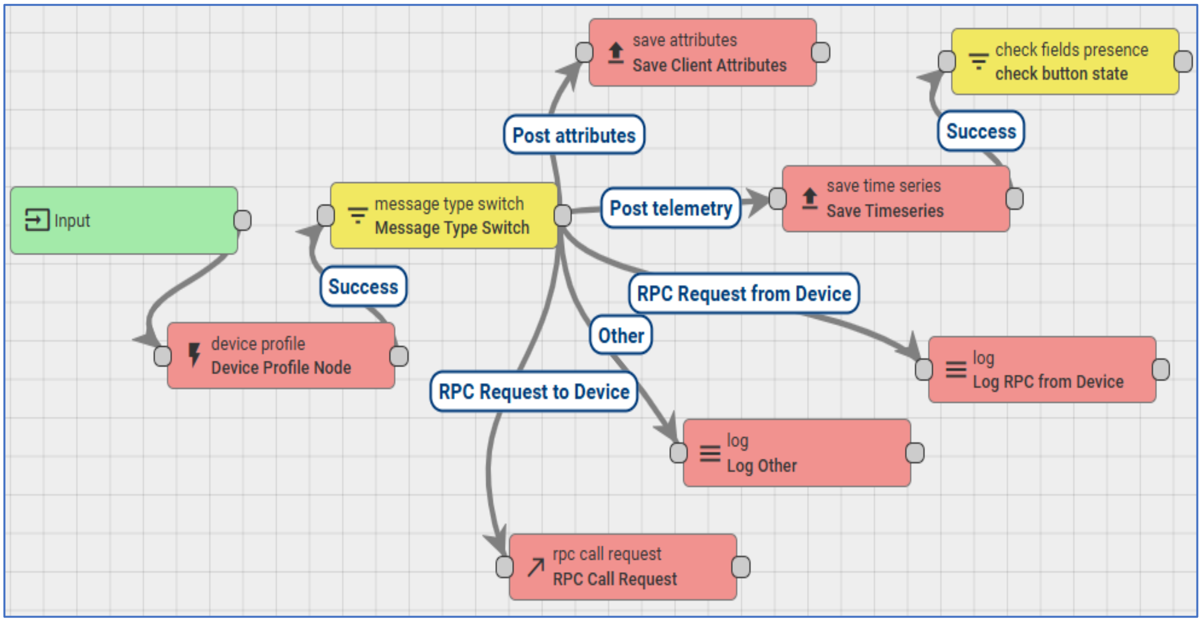


Figure 2.15: Root Rule Chain of the Dashboard

The Root Rule Chain serves as the backend decision engine for the dashboard, processing all incoming MQTT messages from the ESP32 to ensure data is stored or routed correctly.

- **Message Routing:** The core logic relies on a "Message Type Switch" node, which inspects every data and directs it down one of five specific paths based on its content.
- **Data Storage:**
    - **Telemetry Path:** Production data (e.g., total_count, damage_dash) is routed to the "Save Timeseries" node, enabling the dashboard to display historical trends and line charts.
    - **Attribute Path:** Status updates (e.g., boot_time, active_assignee) are sent to the "Save Client Attributes" node, ensuring the device's current state is always up-to-date.
- **Remote Control (RPC):** The chain includes a dedicated "RPC Call Request" path. This handles two-way communication, intercepting commands from dashboard widgets (like changing the Moulds count) and routing telemetry data from the device (like if valid count proceed, date, hourly updating).

Step 5: User Management - Web

To grant access to supervisors without giving them admin rights:

1. **Customer Management:** Created a customer named as "Customer_01" (Ex: In here, SIL moulding section is the customer)
2. **Add Users:** Assign users to that customer. (In this community edition only few users can add)

   - Navigated to Customers → Manage Users.
   - Added ######@samsonint.com (Supervisor) with "Read/Write" permissions for the dashboard.
   - This allowed the supervisor to log in via the "ThingsBoard Live" mobile app and view only their relevant machines.

3. **Assign Device and Assets:** The devices (E-25, E-69, E-103) were assigned to this Customer group.
4. **Dashboard Sharing:** The created dashboard was shared with the customer group.

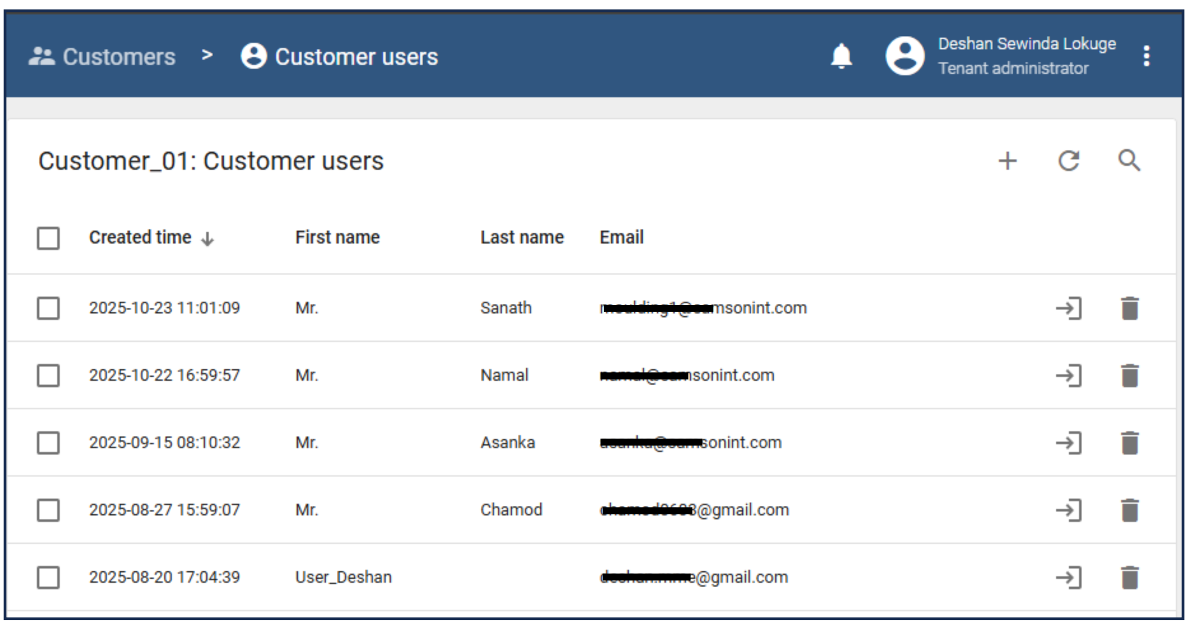


Figure 2.16: Created Customers for the Moulding Section

Step 6: User Management - Mobile

Figure 2.17: ThingsBoard Live Mobile App [2]

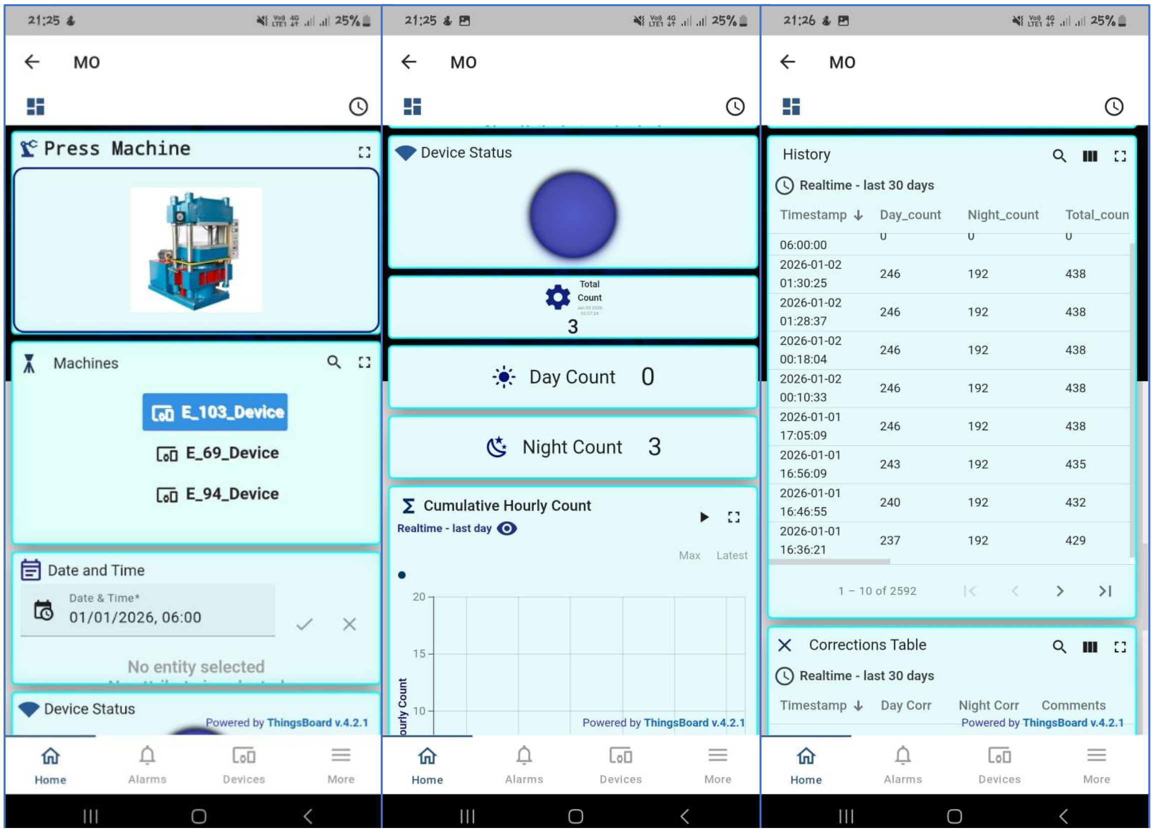


Figure 2.18: Mobile View of the Dashboard

To enhance the system's accessibility for floor supervisors and foremen without incurring additional development overhead, utilized the official **"ThingBoard Live"** mobile application (Available on Google Play and App Store) rather than building a custom solution from scratch. This approach doesn't need coding capabilities, allowing us to bypass the complex Flutter development environment and bundle configuration (with limitation). Implementation was immediate: supervisors and foremen simply downloaded the pre-built app and authenticated using the demo.thingsboard.io endpoint along with their specific user credentials (e.g., email and the password).

Once logged in, the application automatically rendered the existing web dashboards into a mobile layout and one user choose the particular dashboard (Ex: Moulding Section), the widgets are vertically visible for better readability on smaller screens (This mobile view can arrange by the official ThingsBoard web). This provided instant, secure access to real-time count and the information about the machine directly from the factory floor, ensuring that critical production data is available to management even when they are away from the main control terminal.

2.1.6 Further implementations

1. **Implement Over-The-Air (OTA) Updates:** This is already Configured. The firmware development requires a USB Cable and physical connection to the device. Set up the SSID to connect each device on the same network to update the code.

```
; OTA Upload Environment
[env:esp32dev_ota]
platform = espressif32@6.8.1
board = esp32dev
framework = arduino
monitor_speed = 115200
upload_protocol = espota
upload_port = esp32-e-103-production.local   ; Use mDNS name
upload_flags =
    --auth=deshanls
    --port=3232
```

Figure 2.19: OTA Update Configurations

2. **Sample Updating Tables:** This will uniquely show what type of update is done when changing the count values separately.
3. **Cumulative Hourly Count:** Previously, this was a bar chart displaying only 12 hours of data. It has now been updated to a 24-hour line chart. This change was made due to request of supervisors, who needed to monitor regular patterns and identify exactly what time production is stopped.

2.1.7 Future Expansions

1. **Migration to Professional IoT Infrastructure:** The current system relies on the ThingsBoard - demo version which has limitations. For full-scale deployment, the system should migrate to a ThingBoard Cloud version (monthly subscription based) or the ThingsBoard Professional Edition (PE). This will allow for:

    - Unlimited data retention (currently limited in the demo).
    - White-labeling the dashboard with the company logo.
    - Complex Rule Chains that were previously restricted in the demo version.

2. **Transition to Custom PCBs (Printed Circuit Boards):** To scale from a few machines to the entire factory floor, the circuit should be converted into a custom PCB design.

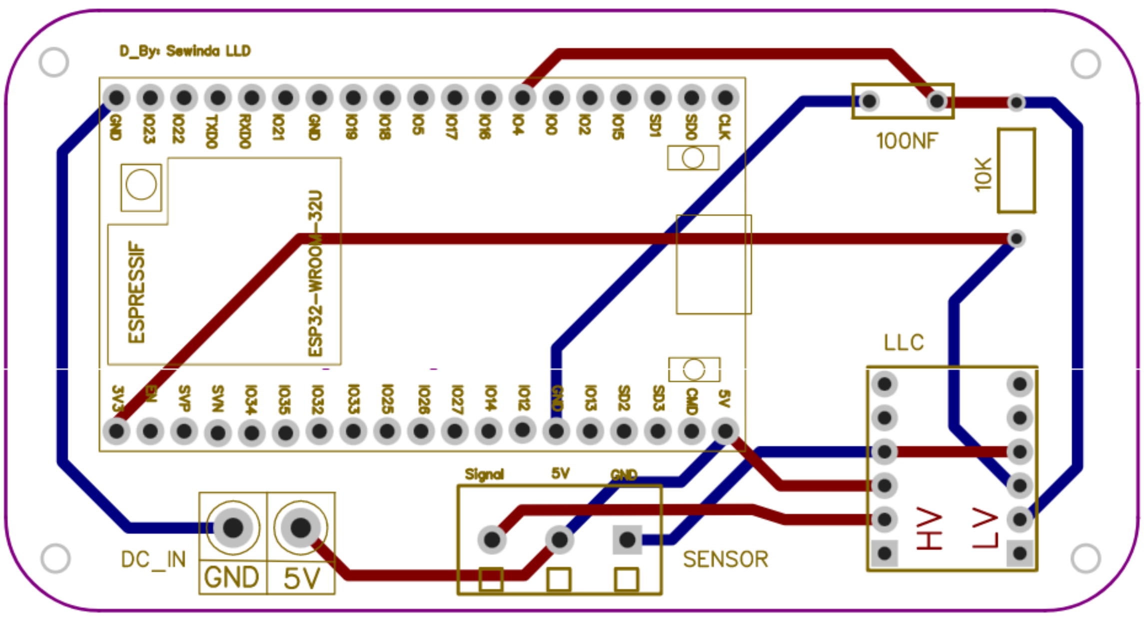


Figure 2.20: Common PCB Design for the Press Machines

The above image shows the designed common PCB for the machines (Based on the machine requirements this can be slightly different.) This will,

- Eliminate loose wires and green terminal connectors which can loosen due to machine vibration.
- Reduce assembly time for new units.

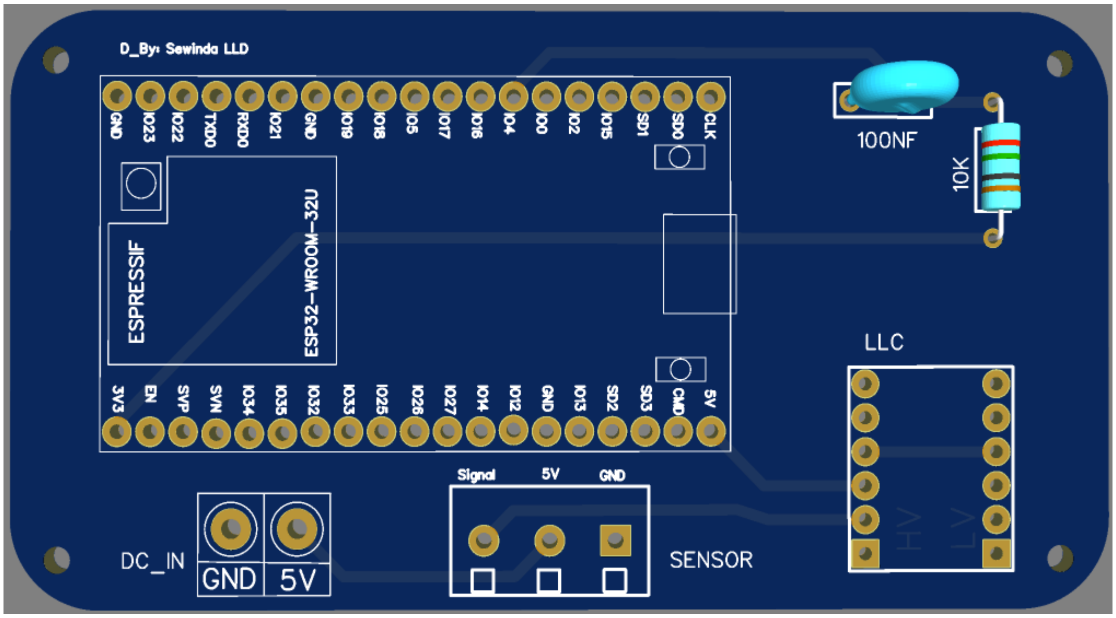


Figure 2.21: 2D View of the Circuit

3. **Scaling for the other press machines:** Same configurations can do some other similar machines by adding new devices (both hardware and the dashboard end setups).

2.1.8 Practical Challenges and Engineering Solutions

1. Problem→ Electromagnetic Interference (EMI) and Wi-Fi Instability: During the installation on the E-69 machine, the ESP32 experienced frequent Wi-Fi disconnections and random reboots. The initial installation placed the plastic IP enclosure *inside* the machine's main electrical cabinet. This cabinet contained large high-current contactors and heavy motors.

- Cause 1 (Faraday Cage): The metal cabinet blocked the Wi-Fi signal.
- Cause 2 (Inductive Noise): The switching of large motors generated significant

Solution: Moved the device enclosure outside the machine's electrical cabinet, mounting it on the external frame. And added a 100nF capacitor across the 3.3V/GND rails of the ESP32 to filter out high-frequency noise spikes. Then the connection stability improved immediately, and the device remained online consistently.

2. Problem→ Signal Bouncing and False Counts (Relay Logic): The press machines are heavy hydraulic machines. When the press closes, it causes significant physical vibration (bumping: to remove the air inside the rubber). The mechanical relays (E-69) contacts would sometimes bounce (briefly disconnect and reconnect) due to this impact, causing the ESP32 to register 2 or 3 counts for a single cycle.

Solution: Hardware adjustment was done by adding filters. For E-69 press machine take the valid cycle by the buzzer signal. If one production is done buzzer will activate based on that take the valid count. (Old machines). To take the valid cycle more accurately implemented code-based solutions,

- Software Debounce: Implemented a "Dead Time" logic in the code.
- Valid cycle durations based on the operation time telemetry.
- Use input pull up resistors in the esp32 board.

3. Problem→Initial moulds_Day/Night Setting Problem: Foremen sometimes forget to update the operation moulds at the start of the shift (6 AM). When this happens, the system incorrectly uses the previous day's mould configuration, resulting in wrong counts.

Solution: A separate variable named day/night_cycles was introduced. This tracks the machine cycles operated directly. Consequently, if any changes are made to the moulds later, the system automatically multiplies the day/night cycle count by the updated mould count to generate the exact production.

4. Problem→Raw Rubber Heating Cycles (Validation Logic on E-94): On the E-94 press, operators sometimes relocate the same product(local) again due to the issues of the compound same production cycle time. (to heat up the raw rubber). The sensor takes this cycle also as a valid cycle and lead to miscount.

Solution: A Validation Button has been installed. This is a physical push-button on the operator panel connected to the edge device, which should be pressed if such a scenario occurs. The system is designed to validate this button press also with the operation time settings to ensure better accuracy.

5. Problem→ThingsBoard demo version limitations: The Root Rule Chain serves as the backend decision engine for the dashboard, processing all incoming MQTT messages from the ESP32 to ensure data is stored or routed correctly. However, complex logics such as multiplying moulds_Day/Night count by actual cycles for a specific day cannot be performed there because the necessary processing node is missing. The Rule Chain simply visualizes the raw telemetry sent from the edge device and performs limited functions to server attributes (e.g., Last Timestamp, Active Status)

Solution: The system retrieves the moulds_Day/Night value from the dashboard, calculates the actual count on the edge device, and sends the result back to the dashboard (Dashboard → Edge Device → Dashboard). However, this bidirectional process significantly increases rule engine executions and telemetry usage. Consequently, the monthly free execution limits of the demo account are exhausted very quickly.

6. Problem→API Usage Limitation: For free users there are limitations for rule engine execution, user adding, devices creating, telemetry requests, customers creating, scripts executions. If the limit exceeds telemetries are dropped.

Solution: Limit the number of devices for a one tenant administrator.

2.2 Design a Ring Counting System for the JSR Packaging Unit.

This project was done by myself and the other two mechanical engineering interns to improve the efficiency of jar sealing ring counting (JSR Counting) and the packing.

2.2.1 Problem Identification and Initial Learning Phase

The manual ring counting process was slow, inefficient and often inaccurate which led to production delays. To overcome this problem, we decided to use machine learning and computer vision to develop an automated counting system. The goal was to design a solution that can detect and count rings in real time using a camera and a trained object detection model.

As As a one of team member, I participated in learning and supporting the early stage of this project. To build a solid foundation, we studied the basics of object detection and how machine learning models can be trained to recognize specific objects. We explored the YOLO (You Only Look Once) model due to its high accuracy and speed for real-time detection. We focused on learning about YOLOv8 version architecture, how it processes images and why it is suitable for tasks like ring counting for industrial setting. After understanding the theory, we moved on to practical work. We began preparing the dataset by collecting sample images of JSR rings from different angles, lighting conditions, and backgrounds. We learned how to annotate the images accurately by drawing bounding boxes around each ring since the proper data preparation was crucial for industry related implementations.

Figure 2.22: Manual Counting of Rings

2.2.2 Model Training, Improvement, and Hardware Support

Once the dataset was ready, we installed the required software libraries and set up the training environment. This included configuring the YOLOv8 model and ensuring the system can handle training tasks efficiently. We trained the first version of our model using the annotated dataset. During this stage, we learned how to monitor training progress and evaluate the model using performance metrics such as precision. Although the results from the first model were not perfect, they successfully demonstrated that the approach was feasible. The model was able to detect and count rings, although with some inaccuracies that could be improved by refining the dataset and adjusting training parameters. Through this experience, we gained a clear understanding of how data quality directly affects model performance and how variations in lighting, background, and ring position can influence detection accuracy. Overall, this provided a strong foundation for the project. We learned the complete workflow of developing an object detection system from identifying the problem and collecting data to training and evaluating a model. It also showed the importance of automation in improving accuracy and efficiency in industrial tasks like ring counting.

Then we focused to improve the system performance and moving closer to real-time implementation. The first step was to enhance the dataset used for training the YOLOv8 model. More labeled samples of rings were added to increase dataset diversity and prevent underfitting. This helped to retrain the model to recognize rings under different lighting conditions, orientations and background variations which are crucial for reliable performance in real-world scenarios. After expanding the dataset, the model was trained and retrained several times to improve detection accuracy. During this process, issues such as false positives and missed detections were observed. To address these, various data techniques were applied, including changes in brightness, image rotation, and scaling. These augmentations helped the model generalize better and become more robust to changes in camera angles and the lighting conditions.

2.2.3 System Integration, Testing, and Evaluation

Once the accuracy reached an acceptable level, the model was tested on real video streams instead of static images. This marked a major step toward integrating the system into a live environment. A camera feed was connected to the model for initial trials, which allowed us to observe how well it could detect and count moving rings in real time. Through this process, we learned about new challenges that arise in video detection, such as motion blur, shadows and changes in lighting. These insights were valuable for further improving the system's robustness and reliability. In parallel with the software improvements, attention was also given to the hardware setup. Then designed a camera mount that would hold the camera securely above the conveyor system. The main goals of the design were to ensure stability, proper alignment, and adjustability for different camera angles and heights. Initial sketches and CAD drawings of the mount were created using SolidWorks where different mounting positions and structures were analyzed for strength and practicality. This phase was both technical and hands-on. It combined machine learning refinement with design development.

Figure 2.23: Conveyor and Camera Mount

The transition from static image testing to real-time video detection was a significant milestone, as it provided a more realistic understanding of system performance. This stage strengthened the connection between hardware and software, moving the project closer to a fully functional automated ring-counting system.

At this stage of the project, the focus moved from model training toward system integration and testing. The goal was to combine all components of the conveyor, camera, and computer into a single working setup. Layout planning was completed to determine the most efficient placement of each part so that the camera could clearly view the conveyor area where the rings pass. This planning step helped ensure that the overall design would be compact, stable, and practical for real-world operation. Once the layout was finalized, the camera mount design was refined in SolidWorks. The updated model focused on alignment, durability, and ease of adjustment. It was important that the camera could be repositioned easily for calibration while remaining stable during operation. A detailed layout drawing of the entire system was also prepared to show how the mechanical and electronic parts would fit together. During this phase, we studied tolerances, load-bearing limits, and alignment issues that could affect detection accuracy.

Even a slight misalignment of the camera was found to cause large variations in detection results. To test these factors in practice, a prototype of the mount was built, allowing real-world evaluation of strength, stability, and camera positioning accuracy.

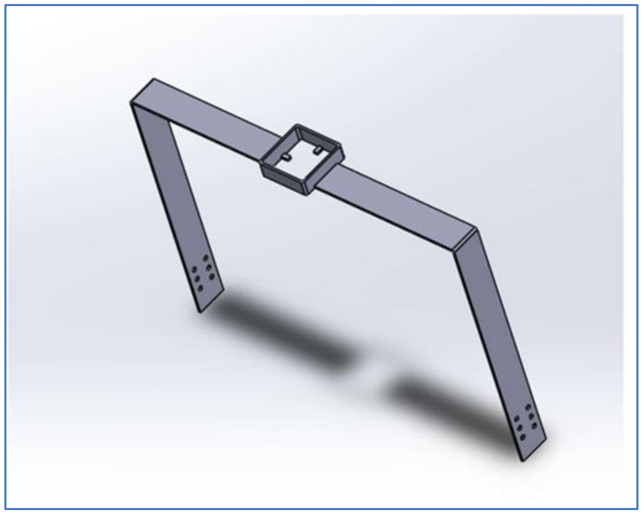


Figure 2.24: Camera Mount

After assembling the mount, the camera was integrated with the setup and adjusted to achieve the best view of the conveyor. Several real-time tests were conducted using the trained YOLOv8 model. Instead of relying solely on the appearance of bounding boxes, a line-crossing method was implemented to ensure more reliable counting. This method helped prevent duplicate detections when the same ring appeared in consecutive frames. The results were evaluated and compared with earlier versions, showing noticeable accuracy gains. Documentation of these tests helped us clearly track progress and validate improvements in the system's detection capability.

The combined setup performed well, demonstrating accurate and efficient ring counting in real time. When compared to manual counting, the automated system showed an efficiency improvement of approximately 15–20%, confirming the success of the project's objectives.

CHAPTER 3

3. Training Experiences – Management

Training and Development Strategies

The SIL has available a training department as a sub-department of the HR department. By this training department, has several training activities for new interns, recruitment members, and employees currently employed. The department prepares a training calendar every year. Accordingly, these training programs are implemented in the institute.

Induction Training

When a new intern is hired by the company, his/her responsibility is one of the functions of this training department until the training period, new intern needed skills, discipline, rules, safety, and company culture, management sides are given by this training department. Sometimes training department allocated them to several sections. (Mixing, Moulding, Calender, etc.)

3.1 Management Practices

Management of a company basically includes the planning, prioritizing and organizing work to achieve the company goals. Management styles are the ways of accomplishing those objectives within the organization. Administrative practices are also play a huge role in this company. As human resource is very important part, the Human Resource Department in the company always try to implement effective workforce management and employee development. The HR Department is interconnected with each and every department in the company. Therefore, the HR department is available several responsibilities for the company. Such as,

- Recruitment process
- Balancing EPF, ETF
- Adjusting the salaries of employees
- Welfare Activities
- Employee performance plans
- Training and development skills and etc.

Number of Employees

The DSI Samson Group employs a total of approximately 10,000 permanent staff. Within this group, Samson International PLC has a permanent workforce of around 450 employees, including staff personnel.

Working hours/ short leaves

At Samson International PLC, the workforce is broadly categorized into two primary groups: Staff and Non-Staff employees. Our employee base is diverse, with both female and male representation across all operational levels. The specific working hours for employees are determined by their respective employment category.

Table 3.1: Working Time Table

Staff employee (both male and female)	7.30 AM to 5.30 PM.
Non- Staff or working employees	Female employees = 7.30 AM. to 5.30 PM. Male employees = Shift 01 6.00 AM to 6.00 PM Shift 02 6.00 PM to 6.00 AM

Also, all employees can take short leaves. But it is also different between staff persons and non-staff person. Normally staff leave can get at 2.00 P.M. or 4.00 PM, also they can get half day at 12.30 PM for once a month and non-staff employees can get 11.00 A.M, or before the evening tea time (3.00 P.M.). And also Training student can get short leave only before the evening tea time.

3.2 Leaves

According to the first and office category, there are two types of leave available in the industry.

Such as laborers' leave and staff leave.

<u>Labors' leaves</u>

Under Sri Lankan labor laws, employees are entitled to a minimum of 14 days of annual leave. However, Samson International PLC goes beyond this basic requirement by adopting a more flexible and employee-friendly leave policy. The company takes into account how long an employee has been with the organization, allowing long-serving staff to earn up to 30 days of annual leave, which reflects its appreciation for loyalty and commitment.

While the standard 14 days of leave is the legal baseline, taking additional time off beyond this is usually considered unpaid leave. Even so, the company understands that life doesn't always fit neatly into rules. In exceptional or compassionate situations, the Human Resources team may grant extra leave days and in many of these cases, employees continue to receive their basic salary without any deductions.

Staff leaves

According to the first and office-grade employment package, staff members are entitled to 14 days of annual leave and 7 days of casual leave.

If Factory doesn't run production process top management give factory leaves for the employees and intern employees. Also, employee can get medical leaves, medical certificate has to submit. Leaves for pregnant mothers (With Salary) – 87 Days (1st children) / 42 Days (2nd Children). Factory not working on these days - Poya day / Special Government Holidays.

Employees' Provident Fund (EPF)

The Employees Provident Fund (EPF) was established under Act No.15 of 1958. This fund purpose is to help employees to save a part of salary every month. The fund can be used when the employee retire from the work. It is a kind of social security to the industrial workers. EPF is the most important type of provident fund to save money. The EPF amount equal to 20% of the employee's total earnings to the fund. The 20% was calculated as, 8% from the monthly salary of employee and 12% contribution from the company. Contribution should be paid on the total earning of the employee. Total earning includes salary, fees, cost of living allowance and holyday. The EPF's account holder can nominate his family members as a nominee.

Employee Trust Fund (ETF)

An employee is entitled to Employee Trust Fund (ETF) from the first day of his/her employment irrespective of whether he/she is permanent, temporary, apprentice, casual or a shift worker. Similarly, employees working on piece rate, contract basis, and work performed basis of any manner are also eligible for membership. Employer has to contribute an amount equal to 3% of the employee's total earnings. This method is done by a contribution of 3% of employee's salaries and no deduction from the employee's salary.

Compensation and welfare Activities

Insurances

All employees are insured by the company when they are permanently joined with the company. If the employee is hospitalized in a private hospital the bill is paid by SIL with the hope of claiming from an insurance company. If an employee becomes permanently disabled, his basic salary will be paid for the rest of his life. Special allowances are paid according to the culture of the company. If any employee is subjected to a surgery within his service time, he is paid Rs 500,000. And also, they are paid Rs. 12,000 annually as

medicine charges. Apart from that the family of the employee is given insurance coverage and free medical check-ups.

Especially medical facilities

Every six months, all employees undergo a free medical examination, and more highlighted employees are referred are further examination and treatment and this will be paid in full by the company. In addition, a special doctor comes once a week, where special needs workers can get free medical care. The six-monthly medical program provides free spectacles to those who need them and is available under loan assistance to employees who require high-quality spectacles.

Funeral

A funeral Fund is run through the HR departments and is common for both staff and employees. One hundred and eighty (150) rupees will be allocated from the salary of each employee once a month for that. Rs. 50000 and other necessary equipment or facilities will be provided at the funeral of an employee's family member. At some point, when an employee die, paid Rs. 100000 in cash and his basic salary is paying to his named family member. But on the special rules.

Special offers

Every year, all employees and staff receive three uniforms at no cost. The company also offers free transportation and complimentary breakfast. Lunch and dinner are provided at a subsidized rate, making meals more affordable. Also, employees are supplied with personal protective equipment (PPE) free of charge, ensuring their safety at work.

Retirement Benefits

Upon retirement, employees do not receive a traditional pension or special retirement bonus. However, they are entitled to their EPF (Employees' Provident Fund) and ETF (Employees' Trust Fund) contributions, along with special gifts and cash awards as a token of appreciation. If a retired employee faces financial difficulties, there are opportunities to return to work on a contract basis. Additionally, retirees may be offered subcontract work with the company, such as tasks like cutting V-straps, shoe soles, or printing, depending on their skills and experience. The company also honors long-term service. When an employee reaches 20 years of service, a special ceremony known as a "service valuation" is held to recognize and celebrate their dedication.

Education Facilities

Special Scholarships are offered to uplift the future of the children of staff and employees. Scholarships up to A/L year are given to students who pass the age of five up to the advanced level years. Special scholarships up to Advanced Level education are given to students who excel in the Scholarship Examination. In addition to, a special function will be held for the students to further encourage the highly successful students under the patronage of the top officials of the company. In addition, leadership training courses are provided free of charge at school and university levels, as well as specialized training offers, will give to university students within the institution.

Organization programs of development of mental health

Samson International PLC places strong emphasis on the mental and emotional well-being of its employees, offering a variety of programs and activities to support a healthy and positive workplace culture. An annual company get-together is held at a five-star hotel, featuring a full day of entertainment, including performances by popular music bands—providing employees with a chance to relax, have fun, and enjoy quality time with colleagues.

Cultural and spiritual activities are also a key part of the company’s calendar. Sinhala and Tamil New Year celebrations are organized annually, and on special Poya days, events such as Dana (almsgiving), Sheela (observance of precepts), meditation sessions, and dansals (food offerings) are arranged to foster peace and mindfulness.

The company also encourages community involvement, organizing public service activities like cleaning hospitals and maintaining elder care homes. Employees are actively involved in these initiatives, promoting social responsibility and teamwork. To promote health and team spirit, the company hosts inter-division sports events within the DSI Group, as well as friendly competitions with external companies, giving staff the opportunity to showcase their athletic talents. In addition, leadership development workshops are regularly held, featuring guest speakers and expert trainers, aiming to build skills, boost motivation, and enhance overall employee performance.

3.3 Industrial Safety

"SAFETY FIRST" is a common statement that can be seen in any industry. SIL has safety departments. This department will carry out the following activities for the safety of the company and its employees.

When an unsafe condition occurred, the safety departments follow five steps to avoid those unsafe situations.

1. Eliminate
2. Substitute
3. Engineering controls
4. Administrative controls
5. PPE (Personal Protective Equipment)

Samson International PLC (SIL) operates within the chemical manufacturing industry, which naturally involves exposure to chemical substances and physical hazards. Employees may face potential risks from various sources, including machinery, fire, high-temperature surfaces, slippery floors, and moving machine parts.

Accidents and safety risks at SIL are most likely to occur in the following areas:

- The chemical weighing and mixing section, where workers are exposed to hazardous and toxic chemicals, including chemical dust.
- Areas with high-temperature equipment such as root cure machines, hydraulic presses, sheeting sections, hot molding stations, and milling machines, which pose burn and heat-related risks.
- Wet or slippery work floors, which increase the chance of slip-and-fall incidents.

3.3.1 Personal Protective Equipment

Personal protective equipment is supplied to employees by SIL. Employees are strictly advised to wear personal protective equipment. Safety shoes are supplied to all employees of SIL. Overalls and safety helmets are given to all technicians and electricians. It is a must to wear this safety equipment when they are on duty. Because lots of chemicals are used in the compounding section, the employees of that section are given gumboots, aprons, and wiser. Listed below are the PPE devices used in SIL.

- ➢ Safety shoes
- ➢ Helmet
- ➢ Safety Goggles
- ➢ Ear Protectors
- ➢ Masks
- ➢ Boots
- ➢ Hand Covers Hand Glove

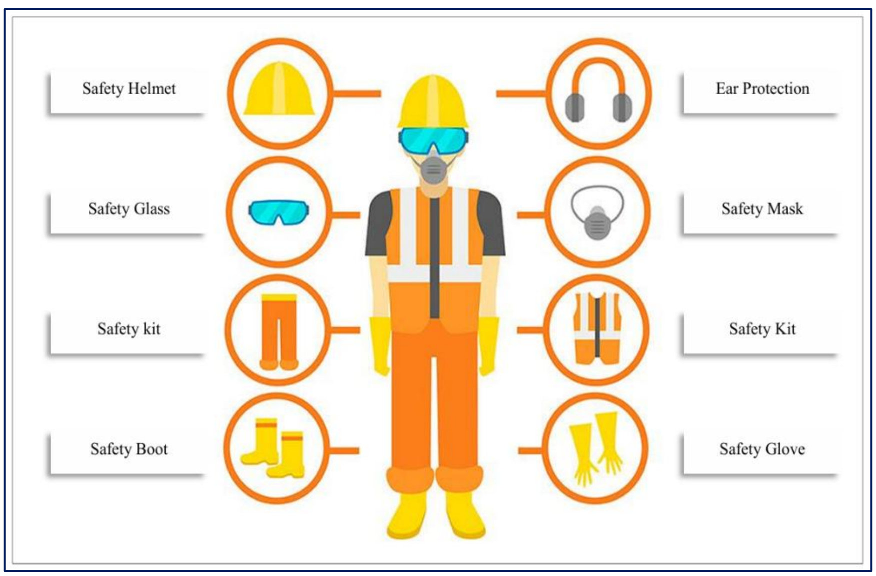


Figure 3.1: Personal protective equipment [4]

Safety helmet

Safety helmets should be worn any place in the Workshop. Even a small bolt drop to the head from a high place may cause serious injuries. Wearing a safety helmet is not sufficient, it should be worn appropriately.

Safety shoes

Toecap prevents damages to fingers in a case of falling high weights on the toe. Oil resistance, the non-slip sole is good protector while working along the slippery oily surface.

Safety goggles and dark glass

Normal safety goggles should be worn with chipping and grinding processes to prevent small particles strike on the eyes. The dark glass should be worn with gas cutting and welding processes to prevent loss of sight due to ultraviolet radiation.

Dust mask

A dust mask is a flexible pad held over the nose and mouth by elastic or rubber straps to protect against dust encountered during construction or cleaning activities.

Safety gloves

Rigger gloves should be worn when working with sharp, corroded, damaged, heavy, etc. objects. It prevents fingers and palms from injuries. Welding gloves should be worn during welding, gas cutting processes, and when handling hot items. The leather material of the gloves absorbs heat, but it is not suitable for handling hot items for a long time.

Ear protectors

The unwanted sound was called noise. Noise-induced hearing damage. The threshold was temporarily raised when the ear was exposed for a length of time to a noise level of about 70-75 db.

Safety overall

Safety overall should not too long or too short. Too long overalls may cause falls, particularly walking by stairways. All buttons of the overall should fasten. Sleeves should be tightly rolled or fully rolled out and put the buttons to avoid damages from sparks during gas cutting processes.

3.3.2 Fire Safety

Any industry can suffer major harm from fire. In rubber industry it is can be very dangerous when fire accident happened. There are therefore adequate safety measures for fire protection. Fire extinguishers need to be accessible and visible in the manufacturing sector. Employees should also be able to recognize the correct extinguisher to relevant accident. Smoking is prohibited in the factory and special area locate in the factory for smoking area. And management of the factory give special training regarding to the fire Safety for the team of the workers. Those workers named as the fire team and they responsible for the fire safety activities for next 6 months. And fire extinguishers and assemble points are located inside the factory. When Fire Alarm is heard, the employees are instructed to do follow the followings.

- Stop the work immediately.
- If the shutdown switch is nearby then shutdown the machine.
- Quickly come out from the plant and come to the assembly point.
- If the main entrance is blocked use the emergency exit way.
- When came to the assembly point look whether people who were working with you also came to the assembly point or not.
- If not then immediately inform to the person who is in charge there.

Figure 3.2: Safety Measures

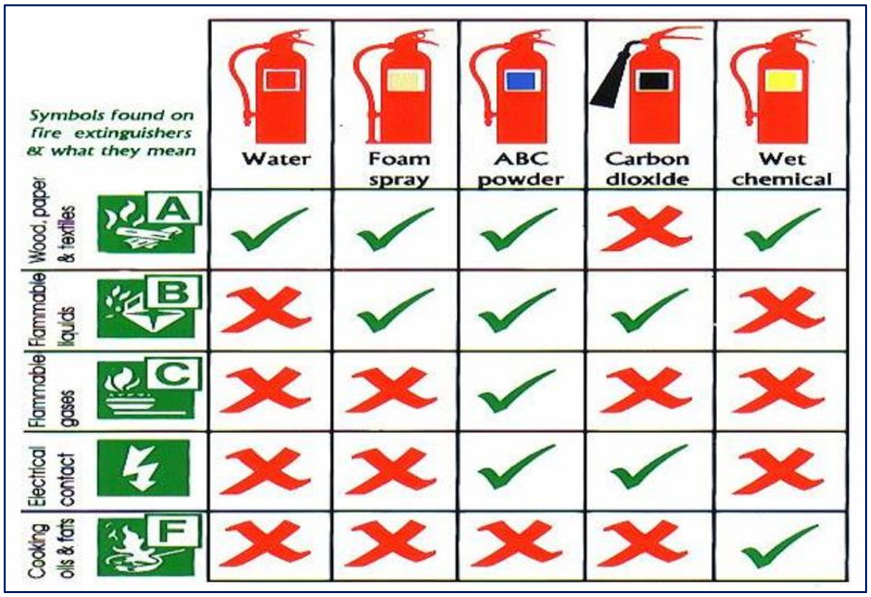


Figure 3.3: Fire Distinguishes

3.3.3 Special Instruction for Machine

- Effective and proper working guards should always be in place when operating machines.
- Standards should be fallow during the calibration of machines and other equipment.
- Shouldn't try to repair the machines while operating, especially the moving parts.
- Always be cautious about the clothes and hands while operating the machines.
- Do not try to stop the machines by holding the moving parts.
- Sufficient light system must be provided during working.
- Workpieces and tools must be clamped on the machine securely.

3.3.4 SIL Safety Rules for Workers

- If workers are working on maintenance, they must wear safety helmets and safety goggles.
- If they are working on a higher level of more than six feet then they must take permission requests.
- Safety shoes must be worn inside the plant.
- Machines must be isolated before starting the maintenance on it.
- All the machines and other instrument should be operated by well-trained authorized persons.
- Prohibited unauthorized activities such as smoking, alcohol, and taking other drugs.
- Proper tools and standard methods must be used in working premises.
- Should clean own workplace every day.

- In hazardous situations individuals must follow the guideline given under the safety procedure.

3.4 Medical and Safety Protocols for Accidents and Incidents

Medical protocols in a rubber factory are essential to ensure prompt and appropriate medical attention in case of accidents or incidents. The following procedures should be in place:

- First Aid Training: Employees should receive basic first aid training to provide immediate assistance to injured colleagues. Designated first aiders should be available on the factory floor during working hours.
- Emergency Medical Services: The rubber factory should have an established emergency medical service that can quickly respond to accidents or incidents within the facility. This may involve having an on-site medical team or coordinating with nearby medical facilities.
- Incident Reporting: Accidents and incidents should be reported immediately to the designated personnel responsible for managing medical emergencies. Proper documentation of the event, injuries sustained, and initial treatment provided is essential for further investigation and follow-up.
- Medical Facilities: The rubber factory should have a designated medical treatment area equipped with essential medical supplies and equipment. In the event of serious injuries, workers should be transported to nearby medical facilities for comprehensive care.
- Evacuation Plans: In the case of major incidents, the rubber factory should have well defined evacuation plans to move employees to safe assembly points or designated medical treatment areas.

Safety protocols play a critical role in preventing accidents and minimizing the impact of incidents in a tire factory. The following safety measures should be implemented:

- Risk Assessments: Regular risk assessments should be conducted to identify potential hazards and address safety concerns. This will help in implementing preventive measures and ensuring a safer work environment.
- Safety Training: All employees should undergo comprehensive safety training that includes safety procedures, emergency response protocols, and the proper use of Personal Protective Equipment (PPE).

- Equipment Inspections: Machinery and equipment should undergo regular inspections to identify and rectify any potential safety issues. Faulty or malfunctioning equipment should be taken out of service until repaired.
- Hazard Signage: The tire factory should have clear and visible hazard signage throughout the facility to warn employees of potential dangers and remind them of safety protocols.
- Emergency Drills: Regular emergency drills should be conducted to familiarize employees with evacuation procedures and emergency response actions.
- Incident Investigation: Every accident or incident should be thoroughly investigated to determine the root cause and prevent similar occurrences in the future.
- Safety Committees: Establishing safety committees can help promote a safety culture and provide a platform for employees to share safety concerns and suggestions.

CHAPTER 4

4. Practice of Professional Standards and Engineering Ethics

Engineering is essential for improving the quality of life, which is why it requires strict adherence to ethical principles and professional standards. During my industrial training at Samson International PLC (SIL), I observed and practiced values such as professionalism standards, integrity, and a deep responsibility toward public welfare. The work environment at SIL promotes accountability and transparency, which are necessary when managing complex manufacturing processes and engineering improvements that impact product safety.

4.1 Professional Standards:

Samson International's engineering and manufacturing practices focus on accuracy, quality, and following strict industry and organizational standards. Qualified engineers at SIL hold impressive credentials for Engineering, Manufacturing, Quality Assuring and R&D. These professionals follow structured procedures to ensure the delivery of high-quality polymer solutions.

Throughout my training, we followed best practices in quality management and process engineering to maintain the integrity of our products. This included regular in-house and overseas laboratory tests to ensure international recognition and safety. SIL has been ISO 9001 certified since 1994, and its PVC products comply with local SLS standards, such as SLS 1206:2000 and 993:2013, reflecting the professional standards supported by the IESL. The engineering team, led by managers with advanced degrees in manufacturing, ensures that all machinery and production lines operate under these rigorous benchmarks

4.2 Ethical Practices in Action:

During my training, I saw first-hand the ethical responsibilities involved in industrial manufacturing. Working with raw materials requires a commitment to public health; for example, SIL strictly follows Standards to ensure that no hazardous chemicals are used in its products. Environmental stewardship is also a core ethical value at the company.

The company also demonstrated its commitment to the community and social responsibility during a very difficult financial year. Despite facing significant economic challenges and reporting its first net loss in 12 years, the company chose to not terminate any of its 458 employees. Instead, SIL supported its workforce by providing food bags and gift vouchers to help them cope with the rising cost of living. This reflects a robust Code of Business Conduct and Ethics, where the welfare of people is prioritized over short-term profits

Furthermore, the economic crisis in Sri Lanka provided several situations that made me rethink the application of ethics in a business setting. Engineering ethics often involve balancing technical efficiency with social impact. One notable ethical decision was SIL's choice to decrease the prices of finished goods already in stock during crisis situation. This was done to remain fair to the market and competitive, even though it resulted in a financial loss for the company.

Additionally, SIL has integrated social welfare into its operational design by establishing a network of trimming centers within local villages. This model provides a vital source of income for vulnerable individuals, such as the elderly and disabled, who are unable to work in a traditional factory setting. By intentionally balancing automation with community-based manual work, the organization demonstrates an ethical commitment to inclusive growth. This approach reflects the professional standard of using engineering solutions not just for technical efficiency, but as a proactive tool for social good and equitable opportunity, ensuring that the production chain directly benefits the surrounding community

CHAPTER 5

5. Environment and Sustainability

Sustainability in engineering involves developing solutions that balance technological advancement, economic efficiency, and environmental responsibility. At Samson International PLC (SIL), sustainability is not just a goal; it is an integral part of the corporate ethos and everyday engineering practices. This chapter reflects on the sustainability initiatives observed during my industrial training, specifically how SIL balances the triple bottom line: Profits, People, and Environment. SIL holds ISO 14001 (Environmental Management) and ISO 50001 (Energy Management) certifications, proving its dedication to the "Planet" pillar of sustainability.

5.1 Environmental Considerations

Energy Efficiency through Green Energy Projects:

SIL has made significant strategic investments in renewable energy to minimize its carbon footprint. The main plant operates a ~750 KW Solar PV system, which generates approximately over 85,000 units of clean electricity per month for the national grid. Additionally, the company utilizes day and night shifts to effectively manage its maximum energy demand and lower overall operational costs. To further reduce dependence on fossil fuels, the company invested in a 3-ton biomass boiler to replace furnace oil for steam generation. [5]

• **Optimized Resource Utilization and 3R Concept:**

The company strictly follows the 3R (Reduce, Reuse, and Recycle) method in managing rubber and water waste. Engineering solutions are designed to recycle reject rubber into value-added products such as road humps, D-bumpers, and floor mats. Furthermore, an in-house wastewater treatment plant with a capacity of 50,000 liters per day ensures that all discharged water meets the standards of the Central Environmental Authority.

• **Ecosystem Protection and Carbon Sequestration:**

SIL actively manages its "Natural Capital" by maintaining a company-owned paddy field and a plantation of trees. These plants are estimated to absorb Carbon Dioxide regularly. To maintain air quality for the community and staff, dust collecting machinery was installed for the mixers in both the rubber and PVC factories.

5.2 Social Considerations [4]

• **Community Integration and Social Well-being:**

SIL is dedicated to serving as a responsible corporate citizen by making a positive impact on the neighboring villages. A unique social initiative is the operation of 45 trimming operation centers, which provide income-generating work for 250 elderly, disabled, or house-bound individuals who cannot work in a traditional factory setting. This model fosters social inclusion and enhances the living standards of vulnerable groups.

• **Employee Support and Retention:**

During the recent economic crisis, SIL demonstrated its ethical commitment by not terminating any of its 458 employees. To support their livelihoods, the company distributed food bags and gift vouchers worth approximately Rs. 10,000 to each employee to help them cope with the high cost of living. This focus on human capital resulted in a high employee retention rate. [5]

• **Knowledge Sharing and Skill Development:**

The training environment at SIL encourages the development of an empowered workforce. I had the opportunity to observe 19 different training programs during the year, covering technical areas like Basic Rubber Technology and safety protocols like Fire Drills. This collaborative environment ensures that sustainable engineering practices are passed down to all levels of the organization.

5.3Economic Considerations

• **Cost Efficiency through Sustainable Technology:**

By investing in solar energy and biomass, SIL has managed to mitigate a substantial portion of its factory electricity and fuel costs. These engineering solutions provide a long-term financial advantage while reducing environmental damage. The company also utilizes natural light through transparent roofing sheets to further lower energy expenses.

• **Local Value Addition and Foreign Exchange Savings:**

SIL contributes to the national economy by using local natural rubber for 60% of its raw material needs, which helps save foreign exchange and supports local plantations. By transforming raw rubber into value-added export products like FSC-certified hot water bottles and sealing rings, the company earned over USD 3.5 million in foreign exchange during the year. [5]

• **Strategic Investment in R&D:**

The company invests over Rs. 40 million annually in Research and Development. This investment allows the engineering team to develop eco-friendly products and re-engineer processes to minimize wastage and stay competitive in the global market. [5]

CHAPTER 6

6. Summary and Conclusions

6.1 Summary

During my 12-week industrial training at Samsons International PLC, I had the opportunity to gain hands-on experience in the field of IoT engineering by enhancing my technical and professional skills. Under the supervision of Mr. Namal Nishantha and Mr. Asanka Edirisinghe, I had a great opportunity to work on industrial IoT project, including optimization of JSR production. I gained and learned expertise in designing IoT systems and dashboards. My exposure to ThingsBoard, including its cloud, demo, and local versions, allowed me to develop custom dashboards, configure device integrations and manage of IoT platforms and their applications.

Working with tools like VS Code, Thingboard Live, Arduino IDE, Easy EDA circuit designer and MQTTBroker services helped me efficiently test and configure IoT systems. I overcame challenges such as debugging Arduino codes, connecting all the device to one network to perform efficient IoT system, troubleshooting circuits and integrating physical devices with cloud platforms, which improved my problem-solving and troubleshooting abilities.

Collaborating with supervisors and machine operators strengthened my communication skills and adaptability. Not only that, collaborating was very helpful to address the industry related practical issues with the designed devices. The experience also underscored the importance of teamwork and continuous learning. Despite initial obstacles and resource limitations, the internship helped me grow personally and professionally. Overall, this internship provided a strong foundation in cloud engineering and IoT, preparing me for future opportunities in the field and equipping me with the skills necessary to succeed in my career.

6.2 Conclusions

The training program at SIL PLC has been an invaluable experience, bridging the gap between academic learning and industrial practices. It provided me with exposure to the practical challenges and solutions involved in maintaining IoT networks using data-driven approaches such as circuit designing and Arduino coding. Some of the key takeaways from my training include:

1. **Technical Skills**: Enhanced skills in IoT systems and studied connecting IoT things all together and resolving system faults, and configuring Thingsboard platform into IoT systems with mobile integrations.
2. **Project Management Skills**: Gained proficiency in managing workflows, financial handling with maintaining proper documentation and completing each sub work load aligned with the project timeline.
3. **Professional Development**: Improved professional communication and demonstration, problem-solving, and teamwork skills by working with industry supervisors and engaging with engineers and production-managers

This training experience has increased my interest IoT and its applications in industry, inspiring me to pursue further research in SCADA system designing with industrial automation. The practical exposure at SIL PLC has given me a strong foundation to contribute effectively to the evolving field of real time production IoT systems and data-driven engineering solutions.

REFERENCES

[1] DSI Samson Group, "Founder – DSI Samson Group." [Online]. Available: https://www.dsi.lk/founder/. [Accessed: Feb. 04, 2026].

[2] ThingsBoard, "ThingsBoard Community Edition," *ThingsBoard*. [Online]. Available: https://thingsboard.io/docs/. [Accessed: Feb. 04, 2026].

[3] Samson International, "Samson International | leading rubber products manufacturer in Sri Lanka." [Online]. Available: https://www.samsonint.com/samson-international-quality. [Accessed: Feb. 04, 2026].

[4] Samson International PLC, *Annual Report 2024/2025*. Galle, Sri Lanka: Samson International PLC, 2025.

[5] "MD0480 - SN04-N 4mm Proximity Sensor NPN NO 10-30VDC 300MA 3-wire Inductive." *Tronic.lk*. [Online]. Available: https://tronic.lk/product/sn04-n-4mm-proximity-sensor-npn-no-10-30vdc-300ma-3-wir?srsltid=AfmBOoqTYDj5g_DgdWoALbpjPXkesvHDHKsZ1FsNSUqT6T5eTdjwrx3a. [Accessed: Feb. 04, 2026].

[6] "220/240V AC Coil DPDT Power Relay with base MY2NJ 8 Pin OMRON | Daraz.lk." *Daraz.lk*. [Online]. Available: https://www.daraz.lk/products/220240v-ac-coil-dpdt-power-relay-with-base-my2nj-8-pin-omron-i188581610.html. [Accessed: Feb. 04, 2026].

LIST OF ABBREVIATIONS

3R - Reduce, Reuse, and Recycle

API - Application Programming Interface

BSCI - Business Social Compliance Initiative

CAD - Computer-Aided Design

DSI - D. Samson Industries

EDA - Electronic Design Automation

EMI - Electromagnetic Interference

EPF - Employees' Provident Fund

ESP - Espressif Systems (Microcontroller)

ETF - Employee Trust Fund

GPIO - General Purpose Input/Output

IoT - Internet of Things

ISO - International Organization for Standardization

JSR - Jar Sealing Rings

MQTT - Message Queuing Telemetry Transport

NCE - National Chamber of Exporters

NPN - Negative-Positive-Negative (Transistor Configuration)

OTA - Over-The-Air

PCB - Printed Circuit Board

PLC - Public Limited Company

PPE - Personal Protective Equipment

PV - Photovoltaic

PVC - Polyvinyl Chloride

RC - Resistor-Capacitor

REACH - Registration, Evaluation, Authorization and Restriction of Chemicals

RPC - Remote Procedure Call

SCADA - Supervisory Control and Data Acquisition

SIL - Samson International

SLS - Sri Lanka Standards

SQL - Structured Query Language

YOLO - You Only Look Once

APPENDICES

Appendix A: E – 103 Press Machine Code with All Payload sending and Error Handling

```
#include <WiFi.h>
#include <PubSubClient.h>
#include <Preferences.h>
#include <time.h>
#include <ArduinoOTA.h>
#include <ESPmDNS.h>
#include <WebServer.h>
// ==== Configuration & Credentials ====
#define WIFI_SSID    "Dialog 4G 152"
#define WIFI_PASSWORD "8A047bED"
#define THINGSBOARD_SERVER "demo.thingsboard.io"
#define THINGSBOARD_PORT   1883
#define TOKEN  "k1Zn4jbPIQyaLIAZ9qAQ"
#define BUTTON_PIN 4
#define ATLEAST 0        // Min additional time (ms)
#define ATMOST 450000     // Max additional time (ms)
#define EXCEED_LIMIT 1080000 // 18 mins (Long idle limit)
#define POST_IDLE_SIGNAL_GAP 15000
WiFiClient espClient;
PubSubClient client(espClient);
Preferences preferences;
WebServer server(80);
// Global Variables
unsigned long boot_timestamp = 0;
int received_moulds_day = 0, received_moulds_night = 0;
int received_op_time = 0;
int received_target = 0;
int received_damage = 0;
String received_day_assignee = "", received_night_assignee = "";
int day_count = 0, night_count = 0, total_count = 0;
int raw_day_count = 0, raw_night_count = 0, raw_total_count = 0;
int hourly_counts[24] = {0};
```

```
int current_hour_count = 0, last_tracked_hour = -1;
bool moulds_day_received = false, moulds_night_received = false, op_time_received = false;
bool last_button_state = HIGH;
unsigned long previous_button_timestamp = 0;
unsigned long last_button_time = 0;
bool in_post_idle_sequence = false;
int post_idle_signal_count = 0;
unsigned long signal_2_timestamp = 0;
String last_reset_date = "";
// Helper: Build JSON Payload
String buildCompleteTelemetryPayload(String key = "", String val = "") {
  String p = "{";
  if (key.length() > 0) p += "\"" + key + "\":" + val + ",";
  p += "\"day_count\":" + String(day_count) + ",\"night_count\":" + String(night_count);
  p += ",\"total_count\":" + String(total_count);
  if (getCurrentHour() != -1) p += ",\"total_for_hour\":" + String(hourly_counts[getCurrentHour()]);
  // Add echo variables
  if (moulds_day_received) p += ",\"moulds_Day_dash\":" + String(received_moulds_day);
  if (moulds_night_received) p += ",\"moulds_Night_dash\":" + String(received_moulds_night);
  if (received_day_assignee.length() > 0) p += ",\"day_assignee_dash\":\"" + received_day_assignee + "\"";
  p += ",\"timestamp\":\"" + getCurrentTime() + "\"}";
  return p;
}
// MQTT Callback
void callback(char* topic, byte* payload, unsigned int length) {
  String incoming = "";
  for (unsigned int i = 0; i < length; i++) incoming += (char)payload[i];
  // 1. Handle RPC Requests
  if (String(topic).indexOf("rpc/request") != -1) {
    if (incoming.indexOf("manualReset") != -1) {
      raw_day_count = 0; raw_night_count = 0; day_count = 0; night_count = 0; total_count = 0;
      saveDataToNVS();
```

```
      client.publish("v1/devices/me/telemetry", buildCompleteTelemetryPayload().c_str());
    }
  }
  // 2. Handle Shared Attributes (Parsing Logic Simplified)
  if (incoming.indexOf("moulds_Day") != -1) {
    // ... Parsing logic ...
    int val = parseValue(incoming, "moulds_Day");
    if (val != received_moulds_day) {
        received_moulds_day = val; moulds_day_received = true;
        recalculateProductionCounts();
        client.publish("v1/devices/me/telemetry",
buildCompleteTelemetryPayload("moulds_Day_dash", String(val)).c_str());
    }
  }
  if (incoming.indexOf("Op_time") != -1) {
    received_op_time = parseValue(incoming, "Op_time");
    op_time_received = true;
    saveDataToNVS();
  }
  if (incoming.indexOf("damage") != -1) {
    int dmg = parseValue(incoming, "damage");
    if (dmg > 0) {
        if (isDayShift()) day_count = max(0, day_count - dmg);
        else night_count = max(0, night_count - dmg);
        total_count = day_count + night_count;
        saveDataToNVS();
        client.publish("v1/devices/me/telemetry", buildCompleteTelemetryPayload("damage_dash",
String(dmg)).c_str());
    }
  }
  // ... Other attributes (assignees, targets, corrections) handled similarly ...
}
// NVS Storage Handling
void saveDataToNVS() {
```

```
  preferences.begin("production", false);
  preferences.putInt("raw_day", raw_day_count);
  preferences.putInt("raw_night", raw_night_count);
  preferences.putInt("day_cnt", day_count);
  preferences.putInt("night_cnt", night_count);
  preferences.putInt("op_time", received_op_time);
  preferences.putInt("m_day", received_moulds_day);
  preferences.putInt("m_night", received_moulds_night);
  preferences.putString("d_assign", received_day_assignee);
  preferences.putULong("prev_ts", previous_button_timestamp);
  preferences.end();
}
void loadDataFromNVS() {
  preferences.begin("production", true);
  raw_day_count = preferences.getInt("raw_day", 0);
  raw_night_count = preferences.getInt("raw_night", 0);
  day_count = preferences.getInt("day_cnt", 0);
  night_count = preferences.getInt("night_cnt", 0);
  received_op_time = preferences.getInt("op_time", 0);
  received_moulds_day = preferences.getInt("m_day", 0);
  // ... Load other variables ...
  previous_button_timestamp = preferences.getULong("prev_ts", 0);
  preferences.end();
}
// Core Logic: Process Signal
void processButtonPress() {
  checkMissedDailyReset();
  unsigned long current_time = millis();

  // Post-Idle Startup Sequence Logic
  if (in_post_idle_sequence) {
    post_idle_signal_count++;
    if (post_idle_signal_count == 1) return; // Signal 1: Loading (Skip)
```

```
  if (post_idle_signal_count == 2) {       // Signal 2: Cycle Start
    signal_2_timestamp = current_time;
    return;
  }
  if (post_idle_signal_count >= 3) {       // Signal 3: First Product
    unsigned long gap = current_time - signal_2_timestamp;
    unsigned long min_time = (received_op_time * 1000) + ATLEAST;
    if (gap < min_time) { post_idle_signal_count--; return; } // Too fast
    in_post_idle_sequence = false;
    previous_button_timestamp = current_time; // Reset baseline
  }
}
// Cycle Time Validation (Noise Rejection)
if (!op_time_received || received_op_time <= 0) return;

if (previous_button_timestamp > 0) {
  unsigned long diff = current_time - previous_button_timestamp;
  unsigned long min_cycle = ATLEAST + (received_op_time * 1000);
  unsigned long max_cycle = ATMOST + (received_op_time * 1000);
  if (diff > EXCEED_LIMIT) {
     in_post_idle_sequence = true;
     post_idle_signal_count = 1;
     return;
  }
  if (diff < min_cycle || diff > max_cycle) return; // Reject Noise
} else {
  previous_button_timestamp = current_time; // First press initialization
}
// Valid Count Processing
if (isDayShift()) {
   raw_day_count++;
   day_count = raw_day_count * received_moulds_day;
} else {
```

```
    raw_night_count++;
    night_count = raw_night_count * received_moulds_night;
  }
  total_count = day_count + night_count;
  // Hourly Update
  int h = getCurrentHour();
  if (h != -1) hourly_counts[h] += (isDayShift() ? received_moulds_day : received_moulds_night);
  previous_button_timestamp = current_time; // Update timestamp only on valid count
  saveDataToNVS();
  client.publish("v1/devices/me/telemetry", buildCompleteTelemetryPayload().c_str());
}
void checkDailyReset() {
  String date = getCurrentDate();
  if (date != last_reset_date && getCurrentHour() >= 6) {
    raw_day_count = 0; raw_night_count = 0;
    day_count = 0; night_count = 0; total_count = 0;
    for(int i=0; i<24; i++) hourly_counts[i] = 0;
    last_reset_date = date;
    saveDataToNVS();
    client.publish("v1/devices/me/telemetry", buildCompleteTelemetryPayload().c_str());
  }
}
// Setup & Loop
void setup() {
  Serial.begin(115200);
  boot_timestamp = millis();
  loadDataFromNVS();

  pinMode(BUTTON_PIN, INPUT_PULLUP);
  // Wi-Fi
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED) delay(500);
```

```
  // OTA & Server
  ArduinoOTA.setHostname("ESP32-Press-Monitor");
  ArduinoOTA.begin();
  configTime(19800, 0, "pool.ntp.org");
  client.setServer(THINGSBOARD_SERVER, THINGSBOARD_PORT);
  client.setCallback(callback);
}
void loop() {
  if (WiFi.status() != WL_CONNECTED) WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  if (!client.connected()) reconnect();
  client.loop();
  ArduinoOTA.handle();
   checkDailyReset();
  / Button Debounce & Trigger
  bool reading = digitalRead(BUTTON_PIN);
  if (reading == LOW && last_button_state == HIGH) {
    unsigned long diff = millis() - last_button_time;
    if (diff > 350) { // Hardware debounce
        // Pre-check min cycle time to avoid double counting
        unsigned long min_cycle = (received_op_time * 1000) + ATLEAST;
        if (in_post_idle_sequence || diff >= min_cycle) {
            processButtonPress();
            last_button_time = millis();
        }
    }
  }
  last_button_state = reading;
  delay(10);
}
// Utility Functions
int getCurrentHour() {
  struct tm timeinfo;
  return getLocalTime(&timeinfo) ? timeinfo.tm_hour : -1;
```

```
}
String getCurrentDate() {
  struct tm timeinfo;
  if(!getLocalTime(&timeinfo)) return "";
  char buf[20]; strftime(buf, sizeof(buf), "%Y-%m-%d", &timeinfo);
  return String(buf);
}
String getCurrentTime() {
  struct tm timeinfo;
  if(!getLocalTime(&timeinfo)) return "";
  char buf[10]; strftime(buf, sizeof(buf), "%H:%M:%S", &timeinfo);
  return String(buf);
}
bool isDayShift() {
  int h = getCurrentHour();
  return (h >= 6 && h < 18);
}
// Helper to extract int from JSON string
int parseValue(String json, String key) {
  int start = json.indexOf(key) + key.length() + 2; // skip ":
  int end = json.indexOf(",", start);
  if (end == -1) end = json.indexOf("}", start);
  return json.substring(start, end).toInt();
}
```